



CLI MASTER & NETWORK ENGINEERING GUIDE

ULTIMATE CISCO CHEATSHEET + +

2026

1 Ultimate Cisco CLI Master Cheatsheet & Network Engineering Guide

Ultimate Cisco CLI Master Cheatsheet & Network Engineering Guide covering IOS commands, routing, switching, OSPF, BGP, VLANs, and real-world troubleshooting.

Warning

Production networks require careful planning and change management.

2 Device Fundamentals

2.1 Device Initialization

```
1 ! Complete factory reset (Nuke everything)
2 Router# delete nvram:startup-config
3 Router# delete flash:vlan.dat
4 Router# erase nvram:
5 Router# erase flash:
6 Router# write erase
7 Router# reload
8
9 ! Partial reset (keep IOS)
10 Router# write erase
11 Router# delete vlan.dat
12 Router# reload
```

Danger Zone

These commands will completely wipe your device. Always have backups!

2.2 Boot Process & ROMMON

```
1 ! Interrupt boot process (during first 60 seconds)
2 ! Press Ctrl+Break or send break signal
3
4 ! ROMMON Commands:
5 rommon 1 > confreg 0x2142      ! Bypass startup config
6 rommon 2 > reset                ! Reboot
7 rommon 3 > dir flash:           ! List files
8 rommon 4 > boot flash:c2900-universalk9-mz.SPA.157-3.M.bin
9 rommon 5 > tftpdnld             ! TFTP download
```

2.3 Password Recovery Procedures

For Routers (ISR G2/ASR):

1. Connect console cable
-

-
2. Power cycle, break to ROMMON
 3. `confreg 0x2142`
 4. `reset`
 5. Skip initial config
 6. `enable`
 7. `copy startup-config running-config`
 8. Change passwords
 9. `config-register 0x2102`
 10. `copy running-config startup-config`
 11. `reload`

For Switches (Catalyst):

1. Power off
 2. Hold Mode button while powering on
 3. Release when LED changes
 4. `flash_init`
 5. `load_helper`
 6. `rename flash:config.text flash:config.text.old`
 7. `boot`
 8. `enable`
 9. `rename flash:config.text.old flash:config.text`
 10. `copy flash:config.text system:running-config`
 11. Change passwords
 12. `copy running-config startup-config`
-

2.4 Hardware Inventory

```
1 ! Show hardware details
2 Router# show inventory
3 Router# show diag
4 Router# show environment
5 Router# show module
6 Router# show idprom backplane
7
8 ! Cisco 2960X/3650/3850 Stack Commands
9 Switch# show switch
10 Switch# show switch detail
11 Switch# show switch neighbors
12 Switch# switch 1 provision ws-c3650-24ps
13 Switch# switch 1 priority 15
14 Switch# switch 2 renumber 3
15 Switch# switch 1 reload
```

2.5 IOS Image Management

```
1 ! Verify current IOS
2 Router# show version
3 Router# show boot
4 Router# show flash:
5 Router# dir flash:
6
7 ! Upgrade IOS via TFTP
8 Router# copy tftp://192.168.1.10/c2900-universalk9-mz.SPA.157-3.M
   .bin flash:
9 Router# boot system flash:c2900-universalk9-mz.SPA.157-3.M.bin
10 Router# copy running-config startup-config
11 Router# reload
12
13 ! Bundle/Install Mode (Cat3K/4K)
14 Switch# archive download-sw /force-reload /overwrite tftp
   ://192.168.1.10/cat3k_caa-universalk9.16.12.04.SPA.bin
15
16 ! Install Mode (IOS XE)
17 Router# request platform software package install switch all file
   tftp://192.168.1.10/asr1000-universalk9.17.03.03.SPA.pkg
```

3 IOS Navigation & Basics

3.1 Command Line Fundamentals

```
1 ! Command Abbreviation (up to unique characters)
2 Router# conf t          ! configure terminal
3 Router# sh run          ! show running-config
```

```

4 Router# sh ip int br      ! show ip interface brief
5 Router# wr                ! write memory (copy run start)
6
7 ! Command History & Editing
8 Ctrl+A                   ! Move to beginning of line
9 Ctrl+E                   ! Move to end of line
10 Ctrl+W                   ! Delete previous word
11 Esc+B                    ! Move back one word
12 Esc+F                    ! Move forward one word
13 Ctrl+R                   ! Redisplay line
14 Ctrl+U                   ! Delete entire line
15 Ctrl+Y                   ! Paste deleted text
16 Tab                      ! Auto-complete
17
18 ! Search in Long Outputs
19 Router# show running-config | include ospf
20 Router# show running-config | section router
21 Router# show running-config | exclude !
22 Router# show running-config | begin interface
23 Router# show running-config | count ospf

```

3.2 Context-Sensitive Help

```

1 Router# ?                ! Show available commands
2 Router# show ?           ! Show available show commands
3 Router# configure ?      ! Show configuration options
4 Router# interface gigabitEthernet 0/0/0 ?
5 Router# show ip route <Tab> ! Auto-complete
6
7 ! Partial Command Completion
8 Router# sh ip ro<Tab>    ! Completes to "show ip route"

```

3.3 User Privilege Levels

```

1 ! Configure privilege levels
2 Router(config)# privilege exec level 5 configure terminal
3 Router(config)# privilege exec level 5 show running-config
4 Router(config)# privilege interface level 5 shutdown
5 Router(config)# username admin privilege 5 secret P@ssw0rd
6
7 ! View privilege levels
8 Router# show privilege
9 Router# show privilege all
10
11 ! Enable secret levels
12 Router(config)# enable secret level 5 SuperSecret123
13 Router# enable 5

```

3.4 Command Aliases

```
1 ! Create custom command aliases
2 Router(config)# alias exec srs show running-config | section
3 Router(config)# alias exec shbr show ip interface brief
4 Router(config)# alias exec routes show ip route
5 Router(config)# alias exec neighbors show cdp neighbors detail
6 Router(config)# alias configure config t
7
8 ! Persistent aliases
9 Router(config)# alias exec backup copy running-config tftp
   ://192.168.1.10/
10 Router# backup R1-config.txt
```

3.5 Command Scripting with TCL

```
1 ! TCL Shell Access
2 Router# tclsh
3 Router(tcl)# puts [exec "show version"]
4 Router(tcl)# foreach i {1 2 3 4 5} { puts "Interface Gi0/$i" }
5 Router(tcl)# exit
6
7 ! One-liner TCL commands
8 Router# tclsh puts [exec "show ip interface brief"]
```

4 File System Operations

4.1 File System Navigation

```
1 ! Explore file system
2 Router# pwd                                ! Present working directory
3 Router# dir                                ! List files
4 Router# dir flash:                          ! List flash contents
5 Router# dir nvram:                          ! List NVRAM contents
6 Router# dir usbflash0:                      ! USB flash (if available)
7 Router# dir sdflash:                        ! SD card (ASR/ISR)
8 Router# dir bootflash:                      ! Boot flash
9 Router# dir system:                         ! System files
10
11 ! File Operations
12 Router# copy source-url destination-url
13 Router# copy running-config flash:backup-config
14 Router# copy flash:config.text tftp://192.168.1.10/
15 Router# delete flash:old-config.text
16 Router# erase flash:                        ! Format flash (DANGER!)
17 Router# mkdir flash:backups
18 Router# rmdir flash:backups
```

4.2 Configuration Management

```
1 ! Configuration Archive
2 Router# archive
3 Router(config-archive)# path flash:backups/$h-config
4 Router(config-archive)# maximum 10
5 Router(config-archive)# time-period 1440
6 Router(config-archive)# write-memory
7
8 ! View archive
9 Router# show archive
10 Router# configure replace flash:backups/R1-config-1
11
12 ! Rollback Configuration
13 Router# configure replace flash:backup-config 10
14 Router# configure revert now
15 Router# configure confirm
16
17 ! Configuration Checkpoints (IOS XE)
18 Router# checkpoint database
19 Router# show checkpoint database
20 Router# rollback checkpoint checkpoint_name
```

4.3 Secure Copy (SCP) Operations

```
1 ! Configure SCP server
2 Router(config)# ip scp server enable
3 Router(config)# username admin secret admin123
4 Router(config)# ip ssh version 2
5
6 ! Copy via SCP
7 Router# copy running-config scp://admin@192.168.1.10/backups/R1-
  config
8 Router# copy scp://admin@192.168.1.10/configs/new-config running-
  config
9
10 ! SCP from Linux to Router
11 # scp R1-config admin@10.0.0.1:flash:
```

4.4 USB Operations

```
1 ! Check USB status
2 Router# show usb
3 Router# dir usbflash0:
4
5 ! Copy to/from USB
6 Router# copy running-config usbflash0:/backup-config
7 Router# copy usbflash0:/new-ios.bin flash:
8
```

```
9 ! Boot from USB
10 Router(config)# boot system usbflash0:c2900-universalk9-mz.SPA
    .157-3.M.bin
```

4.5 Automated Backups

```
1 ! EEM Script for Auto-Backup
2 Router(config)# event manager applet AUTO-BACKUP
3 Router(config-applet)# event timer cron cron-entry "0 2 * * *"
4 Router(config-applet)# action 1.0 cli command "enable"
5 Router(config-applet)# action 2.0 cli command "copy running-
    config tftp://192.168.1.10/backups/${_device_name}-
    $_event_pub_time"
6 Router(config-applet)# action 3.0 syslog msg "Backup completed
    successfully"
```

5 Licensing Management

5.1 License Types & Models

```
1 ! Show license information
2 Router# show license
3 Router# show license feature
4 Router# show license udi
5 Router# show license right-to-use
6 Router# show license tech-support
7
8 ! Smart Licensing (IOS XE)
9 Router# show license status
10 Router# show license authorization
11 Router# license smart register idtoken XXXX-XXXX-XXXX-XXXX
12 Router# license smart reservation request local
13 Router# license smart reservation install file flash:reservation.
    lic
14
15 ! Traditional Licensing (Permanent/Evaluation)
16 Router# show license all
17 Router# license install flash:license.lic
18 Router# license boot level ipbasek9
19 Router# reload
```

5.2 License Operations

```
1 ! Save License to File
2 Router# license save flash:all_licenses.lic
3
4 ! Export/Import License
```

```
5 Router# license export flash:license_pa.lic
6 Router# license import flash:new_license.lic
7
8 ! Evaluation License
9 Router# license accept end user agreement
10 Router# license boot level securityk9
11 Router# reload
12
13 ! Check License Compliance
14 Router# show license violation
```

5.3 Right-to-Use (RTU) Licensing

```
1 ! RTU License Management
2 Router# license right-to-use activate ipservices
3 Router# license right-to-use deactivate ipservices
4 Router# show license right-to-use
5 Router# license right-to-use perpetual ipservices accept
```

5.4 DNA Center Licensing

```
1 ! DNA Center Integration
2 Router# license smart register idtoken XXXX-XXXX-XXXX-XXXX
3 Router# license smart reservation request id DNA-CENTER-ID
4 Router# license smart authorization request
5 Router# license smart reservation install
```

6 Device Management

6.1 System Configuration

```
1 ! Basic Device Setup
2 Router(config)# hostname R1-CORE
3 Router(config)# no ip domain-lookup
4 Router(config)# ip domain-name company.com
5 Router(config)# service password-encryption
6 Router(config)# service timestamps debug datetime msec localtime
   show-timezone
7 Router(config)# service timestamps log datetime msec localtime
   show-timezone
8 Router(config)# logging buffered 16384 debugging
9 Router(config)# logging console critical
10 Router(config)# logging monitor debugging
11 Router(config)# logging host 192.168.100.10
12 Router(config)# logging source-interface Loopback0
13
14 ! Banner Configuration
```

```

15 Router(config)# banner motd ^
16 *****
17 *           AUTHORIZED ACCESS ONLY           *
18 *   This system is the property of Company Inc.   *
19 *   Unauthorized access is prohibited and will be prosecuted. *
20 *****
21 ^
22 Router(config)# banner login ^
23 Please enter your credentials to access this device.
24 Contact Network Operations for assistance.
25 ^
26 Router(config)# banner exec ^
27 Welcome to R1-CORE. You are logged in as $username.
28 Current time: $_timenow
29 ^

```

6.2 NTP Configuration

```

1  ! NTP Server Configuration
2  Router(config)# ntp server 0.pool.ntp.org prefer
3  Router(config)# ntp server 1.pool.ntp.org
4  Router(config)# ntp server 2.pool.ntp.org
5  Router(config)# ntp server 192.168.100.10
6
7  ! NTP Authentication
8  Router(config)# ntp authenticate
9  Router(config)# ntp authentication-key 1 md5 NTP-KEY-123
10 Router(config)# ntp trusted-key 1
11 Router(config)# ntp server 0.pool.ntp.org key 1
12
13 ! NTP Master/Peer
14 Router(config)# ntp master 3
15 Router(config)# ntp peer 192.168.1.2
16 Router(config)# ntp update-calendar
17
18 ! NTP Access Control
19 Router(config)# ntp access-group peer 10
20 Router(config)# ntp access-group serve-only 20
21 Router(config)# ntp access-group serve 30
22 Router(config)# ntp access-group query-only 40
23
24 ! Verification
25 Router# show ntp associations
26 Router# show ntp status
27 Router# show ntp clock
28 Router# show clock detail

```

6.3 DNS Configuration

```

1  ! DNS Server Configuration
2  Router(config)# ip name-server 8.8.8.8
3  Router(config)# ip name-server 8.8.4.4
4  Router(config)# ip name-server 192.168.100.10
5  Router(config)# ip domain-name company.com
6  Router(config)# ip domain-list branch.company.com
7  Router(config)# ip domain-list partner.com
8  Router(config)# ip domain lookup source-interface Loopback0
9
10 ! DNS Host Entries
11 Router(config)# ip host router1.company.com 192.168.1.1
12 Router(config)# ip host switch1 10.0.0.1 10.0.0.2
13 Router(config)# ip host ftp-server 192.168.100.10
14
15 ! DNS Caching
16 Router(config)# ip dns server
17 Router(config)# ip dns cache
18 Router(config)# ip dns spoofing 192.168.1.1
19
20 ! Verification
21 Router# show hosts
22 Router# nslookup router1.company.com
23 Router# ping router1.company.com

```

6.4 Syslog Configuration

```

1  ! Syslog Server Setup
2  Router(config)# logging on
3  Router(config)# logging host 192.168.100.10
4  Router(config)# logging host 192.168.100.11
5  Router(config)# logging trap debugging
6  Router(config)# logging facility local7
7  Router(config)# logging source-interface Loopback0
8  Router(config)# logging origin-id hostname
9  Router(config)# logging sequence-numbers
10 Router(config)# logging discriminator OSPF msg-body includes "
    OSPF"
11
12 ! Local Logging
13 Router(config)# logging buffered 16384 informational
14 Router(config)# logging console warnings
15 Router(config)# logging monitor debugging
16 Router(config)# logging persistent url flash:logs size 4096
17 Router(config)# logging history debugging
18
19 ! Logging Filters
20 Router(config)# logging filter OSPF ROUTER-ID 1.1.1.1
21 Router(config)# logging filter BGP neighbor 192.168.1.2
22

```

```
23 ! Verification
24 Router# show logging
25 Router# show logging history
26 Router# show logging filter
```

6.5 SNMP Configuration

```
1 ! SNMPv2c Configuration
2 Router(config)# snmp-server community RO-COMMUNITY RO 10
3 Router(config)# snmp-server community RW-COMMUNITY RW 20
4 Router(config)# snmp-server host 192.168.100.10 version 2c RO-
  COMMUNITY
5 Router(config)# snmp-server enable traps
6 Router(config)# snmp-server trap-source Loopback0
7 Router(config)# snmp-server queue-length 100
8 Router(config)# snmp-server location "Data Center Rack A1"
9 Router(config)# snmp-server contact "Network Operations
  noc@company.com"
10 Router(config)# snmp-server chassis-id R1-CORE-ASR1001
11
12 ! SNMPv3 Configuration (Secure)
13 Router(config)# snmp-server group ADMIN v3 priv read ALL write
  ALL
14 Router(config)# snmp-server user admin1 ADMIN v3 auth sha
  AuthPass123 priv aes 128 PrivPass123
15 Router(config)# snmp-server host 192.168.100.10 version 3 priv
  admin1
16 Router(config)# snmp-server enable traps snmp authentication
  linkdown linkup coldstart warmstart
17 Router(config)# snmp-server enable traps config
18 Router(config)# snmp-server enable traps entity
19 Router(config)# snmp-server enable traps cpu threshold
20 Router(config)# snmp-server enable traps memory threshold
21 Router(config)# snmp-server enable traps ospf
22 Router(config)# snmp-server enable traps bgp
23
24 ! SNMP Views
25 Router(config)# snmp-server view ALL iso included
26 Router(config)# snmp-server view RESTRICTED system included
27 Router(config)# snmp-server view RESTRICTED interfaces excluded
28
29 ! Verification
30 Router# show snmp
31 Router# show snmp user
32 Router# show snmp group
33 Router# show snmp host
```

7 Security Fundamentals

7.1 Password Management

```
1  ! Password Configuration
2  Router(config)# enable secret SuperSecret123!
3  Router(config)# enable algorithm-type scrypt secret
   EvenMoreSecret456!
4  Router(config)# username admin privilege 15 secret P@ssw0rd123!
5  Router(config)# username operator privilege 5 secret Op3rator!
6  Router(config)# service password-encryption
7  Router(config)# security passwords min-length 10
8  Router(config)# login block-for 300 attempts 3 within 60
9  Router(config)# login delay 2
10 Router(config)# login on-failure log
11 Router(config)# login on-success log
12
13 ! Line Password Configuration
14 Router(config-line)# password Cons0leP@ss
15 Router(config-line)# login local
16 Router(config-line)# exec-timeout 5 0
17 Router(config-line)# absolute-timeout 30
18 Router(config-line)# session-timeout 30
19 Router(config-line)# logout-warning 5
20
21 ! Password Recovery Prevention
22 Router(config)# no service password-recovery
23 Router(config)# enable secret recovery SecretRecovery789!
```

7.2 SSH Hardening

```
1  ! SSH Configuration
2  Router(config)# ip ssh version 2
3  Router(config)# ip ssh time-out 60
4  Router(config)# ip ssh authentication-retries 2
5  Router(config)# ip ssh maxstartups 3
6  Router(config)# ip ssh logging events
7  Router(config)# ip ssh server algorithm mac hmac-sha2-256 hmac-
   sha2-512
8  Router(config)# ip ssh server algorithm encryption aes128-ctr
   aes192-ctr aes256-ctr
9  Router(config)# ip ssh server algorithm hostkey rsa-sha2-256 rsa-
   sha2-512
10 Router(config)# ip ssh server disable-deprecated-version 1.99
11 Router(config)# ip ssh server max-sessions-per-connection 10
12
13 ! SSH Key-Based Authentication
14 Router(config)# crypto key generate rsa modulus 4096
15 Router(config)# crypto key generate dsa modulus 2048
```

```

16 Router(config)# crypto key generate ecdsa curve 256
17 Router(config)# ip ssh pubkey-chain
18 Router(config-ssh-pubkey)# username admin
19 Router(config-ssh-pubkey)# key-hash ssh-rsa AAAA...==
20 Router(config-ssh-pubkey)# exit
21
22 ! SSH Source Interface
23 Router(config)# ip ssh source-interface Loopback0
24 Router(config)# ip ssh vrf MGMT
25
26 ! Verification
27 Router# show ip ssh
28 Router# show ssh
29 Router# show crypto key mypubkey rsa

```

7.3 TCP/UDP Small Services

```

1 ! Disable Unnecessary Services
2 Router(config)# no service tcp-small-servers
3 Router(config)# no service udp-small-servers
4 Router(config)# no service finger
5 Router(config)# no ip bootp server
6 Router(config)# no ip http server
7 Router(config)# no ip http secure-server
8 Router(config)# no ip source-route
9 Router(config)# no ip gratuitous-arps
10 Router(config)# no cdp run ! Disable globally
11 Router(config)# no lldp run ! Disable globally
12 Router(config)# no ip proxy-arp
13 Router(config)# no ip unreachable
14 Router(config)# no ip redirects
15 Router(config)# no ip mask-reply
16
17 ! Per-Interface Services
18 Router(config-if)# no cdp enable
19 Router(config-if)# no lldp transmit
20 Router(config-if)# no lldp receive
21 Router(config-if)# no ip directed-broadcast
22 Router(config-if)# no ip unreachable
23 Router(config-if)# no ip proxy-arp

```

7.4 Control Plane Policing (CoPP)

```

1 ! CoPP Configuration
2 Router(config)# class-map match-any COPP-CRITICAL
3 Router(config-cmap)# match access-group 110
4 Router(config-cmap)# exit
5
6 Router(config)# class-map match-any COPP-IMPORTANT

```

```

7 Router(config-cmap)# match access-group 120
8 Router(config-cmap)# exit
9
10 Router(config)# policy-map COPP-POLICY
11 Router(config-pmap)# class COPP-CRITICAL
12 Router(config-pmap-c)# police 8000 conform-action transmit exceed
    -action drop
13 Router(config-pmap-c)# exit
14 Router(config-pmap)# class COPP-IMPORTANT
15 Router(config-pmap-c)# police 40000 conform-action transmit
    exceed-action drop
16 Router(config-pmap-c)# exit
17 Router(config-pmap)# class class-default
18 Router(config-pmap-c)# police 10000 conform-action transmit
    exceed-action drop
19 Router(config-pmap-c)# exit
20 Router(config-pmap)# exit
21
22 Router(config)# control-plane
23 Router(config-cp)# service-policy input COPP-POLICY
24 Router(config-cp)# exit
25
26 ! ACLs for CoPP
27 Router(config)# access-list 110 permit tcp any any eq 22
28 Router(config)# access-list 110 permit tcp any any eq 23
29 Router(config)# access-list 120 permit ospf any any
30 Router(config)# access-list 120 permit eigrp any any
31 Router(config)# access-list 120 permit pim any any

```

8 AAA & Access Control

8.1 AAA Basics

```

1 ! Enable AAA
2 Router(config)# aaa new-model
3 Router(config)# aaa authentication login default local
4 Router(config)# aaa authentication enable default enable
5 Router(config)# aaa authorization exec default local
6 Router(config)# aaa accounting exec default start-stop group
    radius
7
8 ! Local User Database
9 Router(config)# username admin privilege 15 algorithm-type scrypt
    secret AdminPass123!
10 Router(config)# username operator privilege 5 secret Op3rator!
11 Router(config)# username guest privilege 1 secret GuestAccess!
12
13 ! Enable Password
14 Router(config)# enable secret level 15 SuperSecret!

```

```
15 Router(config)# enable secret level 5 OperatorPass
```

8.2 RADIUS Configuration

```
1 ! RADIUS Server Configuration
2 Router(config)# radius server ISE-PRIMARY
3 Router(config-radius-server)# address ipv4 192.168.100.10 auth-
  port 1812 acct-port 1813
4 Router(config-radius-server)# key RadiusKey123!
5 Router(config-radius-server)# retransmit 3
6 Router(config-radius-server)# timeout 5
7 Router(config-radius-server)# exit
8
9 Router(config)# radius server ISE-SECONDARY
10 Router(config-radius-server)# address ipv4 192.168.100.11 auth-
  port 1812 acct-port 1813
11 Router(config-radius-server)# key RadiusKey456!
12 Router(config-radius-server)# exit
13
14 ! RADIUS Server Group
15 Router(config)# aaa group server radius ISE-GROUP
16 Router(config-sg-radius)# server name ISE-PRIMARY
17 Router(config-sg-radius)# server name ISE-SECONDARY
18 Router(config-sg-radius)# exit
19
20 ! AAA with RADIUS
21 Router(config)# aaa authentication login default group ISE-GROUP
  local
22 Router(config)# aaa authentication enable default group ISE-GROUP
  enable
23 Router(config)# aaa authorization exec default group ISE-GROUP
  local
24 Router(config)# aaa accounting exec default start-stop group ISE-
  GROUP
25 Router(config)# aaa accounting network default start-stop group
  ISE-GROUP
```

8.3 TACACS+ Configuration

```
1 ! TACACS+ Server Configuration
2 Router(config)# tacacs server TACACS-PRIMARY
3 Router(config-server-tacacs)# address ipv4 192.168.100.20
4 Router(config-server-tacacs)# key TacacsKey123!
5 Router(config-server-tacacs)# single-connection
6 Router(config-server-tacacs)# timeout 10
7 Router(config-server-tacacs)# exit
8
9 ! TACACS+ Server Group
10 Router(config)# aaa group server tacacs+ TACACS-GROUP
```

```

11 Router(config-sg-tacacs)# server name TACACS-PRIMARY
12 Router(config-sg-tacacs)# exit
13
14 ! AAA with TACACS+
15 Router(config)# aaa authentication login default group TACACS-
    GROUP local
16 Router(config)# aaa authentication enable default group TACACS-
    GROUP enable
17 Router(config)# aaa authorization commands 15 default group
    TACACS-GROUP local
18 Router(config)# aaa authorization config-commands
19 Router(config)# aaa authorization exec default group TACACS-GROUP
    local
20 Router(config)# aaa accounting commands 15 default start-stop
    group TACACS-GROUP
21 Router(config)# aaa accounting exec default start-stop group
    TACACS-GROUP

```

8.4 Command Authorization

```

1 ! Command Authorization Levels
2 Router(config)# privilege exec level 5 show running-config
3 Router(config)# privilege exec level 5 show interfaces
4 Router(config)# privilege exec level 10 configure terminal
5 Router(config)# privilege exec level 15 reload
6 Router(config)# privilege interface level 5 shutdown
7 Router(config)# privilege interface level 10 ip address
8
9 ! Command Sets
10 Router(config)# parser view ROOT
11 Router(config-view)# secret RootView123!
12 Router(config-view)# commands exec include all show
13 Router(config-view)# commands exec include configure
14 Router(config-view)# commands exec include reload
15 Router(config-view)# exit
16
17 Router(config)# parser view OPERATOR
18 Router(config-view)# secret OperatorView456!
19 Router(config-view)# commands exec include show
20 Router(config-view)# commands exec exclude show running-config
21 Router(config-view)# commands exec include ping
22 Router(config-view)# commands exec include traceroute
23 Router(config-view)# exit

```

8.5 Downloadable ACLs

```

1 ! RADIUS with Downloadable ACLs
2 Router(config)# ip access-list extended DYNACL-WEBSERVER
3 Router(config-ext-nacl)# permit tcp any host 192.168.1.10 eq 80

```

```
4 Router(config-ext-nacl)# permit tcp any host 192.168.1.10 eq 443
5 Router(config-ext-nacl)# deny ip any any
6 Router(config-ext-nacl)# exit
7
8 ! AAA Attribute for ACL
9 Router(config)# aaa authorization network default group ISE-GROUP
```

9 Management Protocols

9.1 HTTP/HTTPS Management

```
1 ! HTTP Server Configuration
2 Router(config)# ip http server
3 Router(config)# ip http port 8080
4 Router(config)# ip http authentication local
5 Router(config)# ip http access-class 10
6 Router(config)# ip http secure-server
7 Router(config)# ip http secure-port 443
8 Router(config)# ip http secure-ciphersuite aes256-cbc-sha1
9 Router(config)# ip http timeout-policy idle 60 life 86400
   requests 100
10
11 ! ACL for HTTP Access
12 Router(config)# access-list 10 permit 192.168.100.0 0.0.0.255
13 Router(config)# access-list 10 deny any
14
15 ! Verification
16 Router# show ip http server status
17 Router# show ip http secure-server status
```

9.2 Netconf/RESTCONF

```
1 ! NETCONF Configuration
2 Router(config)# netconf-yang
3 Router(config)# netconf-yang feature candidate-datastore
4 Router(config)# netconf-yang feature notify
5 Router(config)# netconf-yang feature url
6 Router(config)# netconf-yang ssh port 830
7
8 ! RESTCONF Configuration
9 Router(config)# restconf
10 Router(config)# ip http secure-server
11 Router(config)# restconf data-entries 10000
12
13 ! Verification
14 Router# show netconf-yang sessions
15 Router# show restconf
```

9.3 gRPC/Telemetry

```
1 ! gRPC Configuration
2 Router(config)# grpc
3 Router(config-grpc)# port 57400
4 Router(config-grpc)# no-tls
5 Router(config-grpc)# address-family ipv4
6 Router(config-grpc)# max-request-total 64
7 Router(config-grpc)# max-request-per-user 16
8
9 ! Telemetry Configuration
10 Router(config)# telemetry ietf subscription 101
11 Router(config-telemetry)# encoding encode-kvgpb
12 Router(config-telemetry)# filter xpath /interfaces/interface/
    statistics
13 Router(config-telemetry)# source-address 192.168.1.1
14 Router(config-telemetry)# stream yang-push
15 Router(config-telemetry)# update-policy periodic 500
16 Router(config-telemetry)# receiver ip address 192.168.100.30 5432
    protocol grpc-tcp
```

9.4 NETCONF/YANG Models

```
1 ! YANG Models
2 Router# show platform software yang-management process
3 Router# show yang-operational memory
4 Router# show yang schema
5
6 ! Install YANG Models
7 Router# copy tftp://192.168.1.10/ietf-interfaces@2014-05-08.yang
    flash:
8 Router# install add file flash:ietf-interfaces@2014-05-08.yang
    activate
9
10 ! NETCONF Operations
11 # From Linux with netconf-client
12 # netconf-console --host 10.0.0.1 --port 830 --user admin --
    password pass --get-config
```

10 Ethernet Switching

10.1 MAC Address Table

```
1 ! MAC Address Operations
2 Switch# show mac address-table
3 Switch# show mac address-table dynamic
4 Switch# show mac address-table aging-time
5 Switch# show mac address-table count
```

```

6 Switch# show mac address-table interface gigabitEthernet 1/0/1
7 Switch# show mac address-table vlan 10
8 Switch# show mac address-table address 0050.7966.6800
9
10 ! MAC Table Configuration
11 Switch(config)# mac address-table aging-time 300
12 Switch(config)# mac address-table learning vlan 10
13 Switch(config)# no mac address-table learning vlan 20
14 Switch(config)# mac address-table static 0050.7966.6800 vlan 10
    interface gil/0/1
15 Switch(config)# mac address-table notification change
16 Switch(config)# mac address-table limit maximum 1000 vlan 10
17
18 ! Clear MAC Table
19 Switch# clear mac address-table dynamic
20 Switch# clear mac address-table dynamic interface gil/0/1
21 Switch# clear mac address-table dynamic vlan 10
22 Switch# clear mac address-table dynamic address 0050.7966.6800

```

10.2 Port Configuration

```

1 ! Basic Port Configuration
2 Switch(config)# interface gigabitEthernet 1/0/1
3 Switch(config-if)# description "Connected to Server01"
4 Switch(config-if)# switchport mode access
5 Switch(config-if)# switchport access vlan 10
6 Switch(config-if)# switchport voice vlan 20
7 Switch(config-if)# spanning-tree portfast
8 Switch(config-if)# spanning-tree bpduguard enable
9 Switch(config-if)# no shutdown
10 Switch(config-if)# exit
11
12 ! Speed/Duplex Configuration
13 Switch(config-if)# speed 1000
14 Switch(config-if)# duplex full
15 Switch(config-if)# negotiation auto
16 Switch(config-if)# mtu 9216 ! Jumbo frames
17
18 ! Error Disable Recovery
19 Switch(config)# errdisable recovery cause psecure-violation
20 Switch(config)# errdisable recovery cause bpduguard
21 Switch(config)# errdisable recovery interval 30
22 Switch(config)# errdisable detect cause all
23
24 ! Interface Templates
25 Switch(config)# template USER-PORT
26 Switch(config-template)# switchport mode access
27 Switch(config-template)# switchport access vlan 10
28 Switch(config-template)# spanning-tree portfast
29 Switch(config-template)# spanning-tree bpduguard enable

```

```
30 Switch(config-template)# storm-control broadcast level 10.00
31 Switch(config-template)# service-policy input AUTOQOS-VOIP
32 Switch(config-template)# exit
33
34 Switch(config)# interface range gi1/0/1 - 24
35 Switch(config-if-range)# source template USER-PORT
```

10.3 CDP/LLDP

```
1  ! CDP Configuration
2  Switch(config)# cdp run
3  Switch(config)# cdp timer 60
4  Switch(config)# cdp holdtime 180
5  Switch(config)# cdp advertise-v2
6  Switch(config-if)# cdp enable
7  Switch(config-if)# no cdp enable
8
9  ! LLDP Configuration
10 Switch(config)# lldp run
11 Switch(config)# lldp timer 30
12 Switch(config)# lldp holdtime 120
13 Switch(config)# lldp reinit 2
14 Switch(config-if)# lldp transmit
15 Switch(config-if)# lldp receive
16 Switch(config-if)# lldp med
17
18 ! Verification
19 Switch# show cdp neighbors
20 Switch# show cdp neighbors detail
21 Switch# show cdp traffic
22 Switch# show lldp neighbors
23 Switch# show lldp neighbors detail
24 Switch# show lldp traffic
25 Switch# show lldp interface gi1/0/1
```

10.4 UDLD & Loop Guard

```
1  ! UDLD Configuration
2  Switch(config)# udld enable
3  Switch(config)# udld aggressive
4  Switch(config-if)# udld port aggressive
5  Switch(config-if)# udld disable
6
7  ! Loop Guard
8  Switch(config)# spanning-tree loopguard default
9  Switch(config-if)# spanning-tree guard loop
10 Switch(config-if)# spanning-tree guard root
11
12 ! Verification
```

```
13 Switch# show udld neighbors
14 Switch# show udld gi1/0/1
15 Switch# show spanning-tree inconsistentports
```

11 VLANs & Trunking

11.1 VLAN Configuration

```
1  ! VLAN Creation
2  Switch(config)# vlan 10
3  Switch(config-vlan)# name SALES
4  Switch(config-vlan)# state active
5  Switch(config-vlan)# exit
6
7  Switch(config)# vlan 20
8  Switch(config-vlan)# name ENGINEERING
9  Switch(config-vlan)# exit
10
11 Switch(config)# vlan 99
12 Switch(config-vlan)# name MANAGEMENT
13 Switch(config-vlan)# exit
14
15 ! VLAN Range
16 Switch(config)# vlan 100-200
17 Switch(config-vlan)# name USER-VLANS
18 Switch(config-vlan)# exit
19
20 ! Extended VLANs (1006-4094)
21 Switch(config)# vlan 2000
22 Switch(config-vlan)# name EXTENDED-VLAN
23 Switch(config-vlan)# exit
24
25 ! Private VLANs
26 Switch(config)# vlan 500
27 Switch(config-vlan)# private-vlan primary
28 Switch(config-vlan)# private-vlan association 501-502
29 Switch(config-vlan)# exit
30
31 Switch(config)# vlan 501
32 Switch(config-vlan)# private-vlan isolated
33 Switch(config-vlan)# exit
34
35 Switch(config)# vlan 502
36 Switch(config-vlan)# private-vlan community
37 Switch(config-vlan)# exit
```

11.2 VLAN Trunking

```

1  ! Trunk Configuration
2  Switch(config)# interface gigabitEthernet 1/0/24
3  Switch(config-if)# switchport mode trunk
4  Switch(config-if)# switchport trunk encapsulation dot1q
5  Switch(config-if)# switchport trunk native vlan 99
6  Switch(config-if)# switchport trunk allowed vlan 10,20,30,99
7  Switch(config-if)# switchport trunk allowed vlan add 40-50
8  Switch(config-if)# switchport trunk allowed vlan remove 30
9  Switch(config-if)# switchport trunk pruning vlan 2-1001
10 Switch(config-if)# switchport nonegotiate
11 Switch(config-if)# spanning-tree guard root
12 Switch(config-if)# exit
13
14 ! Dynamic Trunking
15 Switch(config-if)# switchport mode dynamic desirable
16 Switch(config-if)# switchport mode dynamic auto
17
18 ! VLAN Translation
19 Switch(config-if)# switchport vlan mapping 10 110
20 Switch(config-if)# switchport vlan mapping 20 120
21
22 ! Verification
23 Switch# show interfaces trunk
24 Switch# show interfaces gi1/0/24 switchport
25 Switch# show interfaces gi1/0/24 trunk
26 Switch# show vlan
27 Switch# show vlan id 10
28 Switch# show vlan name SALES

```

11.3 VTP Configuration

```

1  ! VTP Setup
2  Switch(config)# vtp domain COMPANY
3  Switch(config)# vtp mode server
4  Switch(config)# vtp password VTP-Pass123!
5  Switch(config)# vtp version 2
6  Switch(config)# vtp pruning
7  Switch(config)# vtp interface Loopback0
8
9  ! VTP Client
10 Switch(config)# vtp mode client
11 Switch(config)# vtp transparent
12
13 ! Verification
14 Switch# show vtp status
15 Switch# show vtp password
16 Switch# show vtp counters
17 Switch# show vlan brief

```

11.4 Voice VLAN

```
1 ! Voice VLAN Configuration
2 Switch(config)# interface gigabitEthernet 1/0/10
3 Switch(config-if)# switchport voice vlan 110
4 Switch(config-if)# switchport priority extend cos 0
5 Switch(config-if)# auto qos voip trust
6 Switch(config-if)# mls qos trust cos
7 Switch(config-if)# spanning-tree portfast
8 Switch(config-if)# spanning-tree bpduguard enable
9
10 ! LLDP-MED for Voice
11 Switch(config)# lldp run
12 Switch(config-if)# lldp med
13 Switch(config-if)# lldp med-tlv-select inventory-management
    network-policy
```

11.5 VLAN ACLs (VACLs)

```
1 ! VLAN Access Maps
2 Switch(config)# vlan access-map BLOCK-SERVER 10
3 Switch(config-access-map)# match ip address SERVER-ACL
4 Switch(config-access-map)# action drop
5 Switch(config-access-map)# exit
6
7 Switch(config)# vlan access-map BLOCK-SERVER 20
8 Switch(config-access-map)# action forward
9 Switch(config-access-map)# exit
10
11 Switch(config)# vlan filter BLOCK-SERVER vlan-list 10
12
13 ! VLAN-based ACL
14 Switch(config)# ip access-list extended SERVER-ACL
15 Switch(config-ext-nacl)# deny tcp any host 192.168.10.10 eq 3389
16 Switch(config-ext-nacl)# permit ip any any
```

12 Spanning Tree Protocol

12.1 STP Basics

```
1 ! STP Mode Selection
2 Switch(config)# spanning-tree mode rapid-pvst
3 Switch(config)# spanning-tree mode mst
4 Switch(config)# spanning-tree mode pvst
5
6 ! Bridge Priority
7 Switch(config)# spanning-tree vlan 1,10,20,30 priority 4096
8 Switch(config)# spanning-tree vlan 40-50 priority 8192
```

```

9
10 ! Root Guard
11 Switch(config)# spanning-tree portfast bpduguard default
12 Switch(config-if)# spanning-tree guard root
13 Switch(config-if)# spanning-tree bpdufilter enable
14 Switch(config-if)# spanning-tree bpduguard enable
15
16 ! Verification
17 Switch# show spanning-tree
18 Switch# show spanning-tree vlan 10
19 Switch# show spanning-tree interface gi1/0/1
20 Switch# show spanning-tree detail
21 Switch# show spanning-tree inconsistentports
22 Switch# show spanning-tree mst configuration

```

12.2 MST Configuration

```

1 ! MST Setup
2 Switch(config)# spanning-tree mst configuration
3 Switch(config-mst)# name REGION1
4 Switch(config-mst)# revision 1
5 Switch(config-mst)# instance 1 vlan 10,20,30
6 Switch(config-mst)# instance 2 vlan 40,50,60
7 Switch(config-mst)# exit
8
9 ! MST Instance Parameters
10 Switch(config)# spanning-tree mst 1 priority 4096
11 Switch(config)# spanning-tree mst 2 priority 8192
12 Switch(config)# spanning-tree mst 0-2 hello-time 2
13 Switch(config)# spanning-tree mst 0-2 forward-time 15
14 Switch(config)# spanning-tree mst 0-2 max-age 20
15
16 ! Verification
17 Switch# show spanning-tree mst
18 Switch# show spanning-tree mst configuration
19 Switch# show spanning-tree mst 1
20 Switch# show spanning-tree mst interface gi1/0/1

```

12.3 STP Timers & Optimization

```

1 ! STP Timers
2 Switch(config)# spanning-tree vlan 1 hello-time 2
3 Switch(config)# spanning-tree vlan 1 forward-time 15
4 Switch(config)# spanning-tree vlan 1 max-age 20
5 Switch(config)# spanning-tree transmit hold-count 6
6
7 ! PortFast & BPDU Guard
8 Switch(config)# spanning-tree portfast default
9 Switch(config)# spanning-tree portfast bpduguard default

```

```

10 Switch(config-if)# spanning-tree portfast
11 Switch(config-if)# spanning-tree portfast trunk
12
13 ! UplinkFast & BackboneFast
14 Switch(config)# spanning-tree uplinkfast
15 Switch(config)# spanning-tree backbonefast
16
17 ! STP Loop Guard
18 Switch(config)# spanning-tree loopguard default
19 Switch(config-if)# spanning-tree guard loop

```

12.4 STP Security Features

```

1 ! BPDU Guard
2 Switch(config)# spanning-tree portfast bpduguard default
3 Switch(config)# errdisable recovery cause bpduguard
4 Switch(config)# errdisable recovery interval 30
5
6 ! BPDU Filter
7 Switch(config)# spanning-tree portfast bpdufilter default
8 Switch(config-if)# spanning-tree bpdufilter enable
9
10 ! Root Guard
11 Switch(config-if)# spanning-tree guard root
12
13 ! TC Guard
14 Switch(config)# spanning-tree tc-guard
15 Switch(config)# spanning-tree tc-filter
16
17 ! Verification
18 Switch# show spanning-tree summary
19 Switch# show spanning-tree detail
20 Switch# debug spanning-tree events

```

13 EtherChannel & LACP

13.1 EtherChannel Configuration

```

1 ! LACP Configuration (Recommended)
2 Switch(config)# interface range gigabitEthernet 1/0/1-2
3 Switch(config-if-range)# channel-group 1 mode active
4 Switch(config-if-range)# channel-protocol lacp
5 Switch(config-if-range)# lacp port-priority 1000
6 Switch(config-if-range)# lacp rate fast
7 Switch(config-if-range)# exit
8
9 ! PAgP Configuration
10 Switch(config-if-range)# channel-group 1 mode desirable

```

```

11 Switch(config-if-range)# channel-protocol pagp
12 Switch(config-if-range)# pagp port-priority 1000
13
14 ! Static EtherChannel
15 Switch(config-if-range)# channel-group 1 mode on
16
17 ! Port-Channel Interface
18 Switch(config)# interface port-channel 1
19 Switch(config-if)# description "Uplink to Core"
20 Switch(config-if)# switchport mode trunk
21 Switch(config-if)# switchport trunk native vlan 99
22 Switch(config-if)# switchport trunk allowed vlan 10,20,30,99
23 Switch(config-if)# lacp system-priority 1000
24 Switch(config-if)# lacp system-id 0050.7966.6800
25 Switch(config-if)# exit

```

13.2 Load Balancing

```

1 ! EtherChannel Load Balancing
2 Switch(config)# port-channel load-balance src-dst-ip
3 Switch(config)# port-channel load-balance src-dst-mac
4 Switch(config)# port-channel load-balance src-dst-port
5 Switch(config)# port-channel load-balance src-ip
6 Switch(config)# port-channel load-balance dst-ip
7 Switch(config)# port-channel load-balance src-mac
8 Switch(config)# port-channel load-balance dst-mac
9
10 ! Per-VLAN Load Balancing
11 Switch(config)# vlan 10
12 Switch(config-vlan)# lacp load-balancing src-dst-ip
13 Switch(config-vlan)# exit
14
15 ! Verification
16 Switch# show etherchannel summary
17 Switch# show etherchannel port-channel
18 Switch# show etherchannel load-balance
19 Switch# show lacp neighbor
20 Switch# show lacp internal
21 Switch# show lacp counters
22 Switch# show pagp neighbor
23 Switch# show pagp internal

```

13.3 Advanced EtherChannel

```

1 ! Cross-Stack EtherChannel (StackWise)
2 Switch(config)# interface port-channel 10
3 Switch(config-if)# stack-port 1/1-1/2
4 Switch(config-if)# switchport mode trunk
5 Switch(config-if)# exit

```

```

6
7 ! VSS EtherChannel
8 Switch(config)# interface port-channel 100
9 Switch(config-if)# vss-port-channel
10 Switch(config-if)# switchport mode trunk
11 Switch(config-if)# exit
12
13 ! Minimum Links
14 Switch(config-if)# lacp min-links 2
15 Switch(config)# port-channel min-links 2
16
17 ! Maximum Links
18 Switch(config-if)# lacp max-bundle 8
19
20 ! Graceful Shutdown
21 Switch(config-if)# lacp graceful-convergence

```

13.4 Troubleshooting EtherChannel

```

1 ! Common Issues & Fixes
2 Switch# show etherchannel detail
3 Switch# show etherchannel inconsistency
4 Switch# debug etherchannel
5 Switch# debug lacp
6 Switch# debug pagp
7
8 ! Clear EtherChannel
9 Switch# clear lacp counters
10 Switch# clear pagp counters
11 Switch(config)# default interface port-channel 1
12
13 ! Manual Intervention
14 Switch(config)# interface gi1/0/1
15 Switch(config-if)# shutdown
16 Switch(config-if)# no channel-group
17 Switch(config-if)# no shutdown
18 Switch(config-if)# channel-group 1 mode active

```

14 IP Addressing & Services

14.1 IP Address Configuration

```

1 ! Basic IP Addressing
2 Router(config)# interface gigabitEthernet 0/0
3 Router(config-if)# ip address 192.168.1.1 255.255.255.0
4 Router(config-if)# ip address 10.0.0.1 255.255.255.0 secondary
5 Router(config-if)# ip directed-broadcast
6 Router(config-if)# ip unreachable

```

```

7 Router(config-if)# ip redirects
8 Router(config-if)# ip proxy-arp
9 Router(config-if)# no shutdown
10
11 ! Loopback Interfaces
12 Router(config)# interface loopback 0
13 Router(config-if)# ip address 1.1.1.1 255.255.255.255
14 Router(config-if)# description "Router ID and Management"
15 Router(config-if)# ip ospf network point-to-point
16
17 ! VLSM Subnetting
18 Router(config-if)# ip address 172.16.1.1 255.255.255.192
19 Router(config-if)# ip address 10.1.1.1 255.255.255.224
20 Router(config-if)# ip address 192.168.1.129 255.255.255.240
21
22 ! IPv6 Addressing
23 Router(config-if)# ipv6 enable
24 Router(config-if)# ipv6 address 2001:db8::1/64
25 Router(config-if)# ipv6 address fe80::1 link-local
26 Router(config-if)# ipv6 address autoconfig

```

14.2 DHCP Services

```

1 ! DHCP Server Configuration
2 Router(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
3 Router(config)# ip dhcp excluded-address 192.168.1
4 Router(config)# ip dhcp excluded-address 192.168.1.254
5 Router(config)# ip dhcp pool LAN-POOL
6 Router(dhcp-config)# network 192.168.1.0 255.255.255.0
7 Router(dhcp-config)# default-router 192.168.1.1
8 Router(dhcp-config)# dns-server 8.8.8.8 8.8.4.4
9 Router(dhcp-config)# domain-name company.com
10 Router(dhcp-config)# lease 8
11 Router(dhcp-config)# option 150 ip 192.168.1.10 ! TFTP for VoIP
12 Router(dhcp-config)# client-identifier 0100.5056.c000.08
13 Router(dhcp-config)# host 192.168.1.100 255.255.255.0
14 Router(dhcp-config)# exit
15
16 ! DHCP Relay
17 Router(config-if)# ip helper-address 192.168.100.10
18 Router(config-if)# ip forward-protocol udp 67
19 Router(config-if)# ip forward-protocol udp 68
20
21 ! DHCP Options
22 Router(config)# ip dhcp ping timeout 100
23 Router(config)# ip dhcp ping packets 2
24 Router(config)# ip dhcp conflict logging
25 Router(config)# ip dhcp snooping

```

14.3 DHCP Snooping

```
1 ! DHCP Snooping Configuration
2 Switch(config)# ip dhcp snooping
3 Switch(config)# ip dhcp snooping vlan 10,20,30
4 Switch(config)# ip dhcp snooping information option
5 Switch(config)# ip dhcp snooping limit rate 100
6 Switch(config)# ip dhcp snooping verify mac-address
7 Switch(config)# ip dhcp snooping database flash:dhcp-snooping.db
8 Switch(config)# ip dhcp snooping database write-delay 300
9
10 ! Trusted/Untrusted Ports
11 Switch(config-if)# ip dhcp snooping trust
12 Switch(config-if)# ip dhcp snooping limit rate 50
13 Switch(config-if)# ip dhcp snooping vlan 10 information option
14
15 ! Verification
16 Switch# show ip dhcp snooping
17 Switch# show ip dhcp snooping binding
18 Switch# show ip dhcp snooping statistics
19 Switch# debug ip dhcp snooping
```

14.4 ARP Configuration

```
1 ! ARP Settings
2 Router(config)# arp timeout 300
3 Router(config)# arp 192.168.1.100 0050.7966.6800 arpa
4 Router(config)# arp proxy disable
5 Router(config)# arp gratuitous ignore
6
7 ! Proxy ARP
8 Router(config-if)# ip proxy-arp
9 Router(config-if)# no ip proxy-arp
10
11 ! ARP Inspection (DAI)
12 Switch(config)# ip arp inspection vlan 10,20
13 Switch(config)# ip arp inspection validate src-mac dst-mac ip
14 Switch(config)# ip arp inspection log-buffer entries 32
15 Switch(config)# ip arp inspection log-buffer logs 1024 interval
    10
16 Switch(config-if)# ip arp inspection trust
17 Switch(config-if)# ip arp inspection limit rate 100
```

15 Static Routing

15.1 Basic Static Routes

```

1 ! Standard Static Routes
2 Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
3 Router(config)# ip route 192.168.3.0 255.255.255.0
   GigabitEthernet0/1 10.0.0.3
4 Router(config)# ip route 192.168.4.0 255.255.255.0 10.0.0.4 50
5
6 ! Default Route
7 Router(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.1
8 Router(config)# ip route 0.0.0.0 0.0.0.0 Dialer1
9
10 ! Summary Routes
11 Router(config)# ip route 172.16.0.0 255.255.0.0 10.0.0.1
12 Router(config)# ip route 10.0.0.0 255.255.0.0 Null0
13
14 ! Floating Static Routes
15 Router(config)# ip route 0.0.0.0 0.0.0.0 192.168.1.1
16 Router(config)# ip route 0.0.0.0 0.0.0.0 192.168.2.1 250

```

15.2 Static Route Tracking

```

1 ! IP SLA Tracking
2 Router(config)# ip sla 1
3 Router(config-ip-sla)# icmp-echo 8.8.8.8 source-interface
   GigabitEthernet0/0
4 Router(config-ip-sla)# timeout 1000
5 Router(config-ip-sla)# frequency 5
6 Router(config-ip-sla)# exit
7
8 Router(config)# ip sla schedule 1 life forever start-time now
9
10 Router(config)# track 10 ip sla 1 reachability
11 Router(config-track)# delay down 10 up 5
12
13 Router(config)# ip route 0.0.0.0 0.0.0.0 192.168.1.1 track 10
14 Router(config)# ip route 0.0.0.0 0.0.0.0 192.168.2.1 250

```

15.3 Policy-Based Routing

```

1 ! PBR Configuration
2 Router(config)# access-list 100 permit tcp any any eq 80
3 Router(config)# access-list 100 permit tcp any any eq 443
4
5 Router(config)# route-map PBR-OUTBOUND permit 10
6 Router(config-route-map)# match ip address 100
7 Router(config-route-map)# set ip next-hop 192.168.1.10
8 Router(config-route-map)# set interface Null0
9 Router(config-route-map)# set ip dscp af41
10 Router(config-route-map)# set ip precedence flash

```

```
11 Router(config-route-map)# exit
12
13 Router(config)# interface GigabitEthernet0/0
14 Router(config-if)# ip policy route-map PBR-OUTBOUND
15
16 ! Global PBR
17 Router(config)# ip local policy route-map PBR-OUTBOUND
```

15.4 VRF-Aware Static Routing

```
1 ! VRF Static Routes
2 Router(config)# ip route vrf CUSTOMER-A 192.168.1.0 255.255.255.0
   10.0.0.2
3 Router(config)# ip route vrf CUSTOMER-B 172.16.0.0 255.255.0.0
   GigabitEthernet0/1.100
4
5 ! VRF Leaking
6 Router(config)# ip route vrf CUSTOMER-A 0.0.0.0 0.0.0.0
   GigabitEthernet0/0 global
7 Router(config)# ip route 10.0.0.0 255.255.255.0 GigabitEthernet0
   /0 vrf CUSTOMER-A
```

16 OSPF

16.1 OSPF Basics

```
1 ! OSPF Process Configuration
2 Router(config)# router ospf 100
3 Router(config-router)# router-id 1.1.1.1
4 Router(config-router)# auto-cost reference-bandwidth 1000
5 Router(config-router)# timers throttle spf 10 100 5000
6 Router(config-router)# timers throttle lsa all 10 100 5000
7 Router(config-router)# timers lsa arrival 1000
8 Router(config-router)# max-lsa 12000
9 Router(config-router)# log-adjacency-changes detail
10 Router(config-router)# passive-interface default
11 Router(config-router)# no passive-interface GigabitEthernet0/0
12 Router(config-router)# network 10.0.0.0 0.255.255.255 area 0
13 Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
14 Router(config-router)# exit
15
16 ! Interface Configuration
17 Router(config-if)# ip ospf 100 area 0
18 Router(config-if)# ip ospf hello-interval 10
19 Router(config-if)# ip ospf dead-interval 40
20 Router(config-if)# ip ospf priority 100
21 Router(config-if)# ip ospf cost 10
22 Router(config-if)# ip ospf network point-to-point
```

```
23 Router(config-if)# ip ospf mtu-ignore
24 Router(config-if)# ip ospf authentication message-digest
25 Router(config-if)# ip ospf message-digest-key 1 md5 OSPF-KEY
```

16.2 OSPF Area Types

```
1 ! Multi-Area OSPF
2 Router(config-router)# router ospf 100
3 Router(config-router)# area 1 stub
4 Router(config-router)# area 1 default-cost 100
5 Router(config-router)# area 2 stub no-summary
6 Router(config-router)# area 3 nssa
7 Router(config-router)# area 3 nssa default-information-originate
8 Router(config-router)# area 3 nssa no-redistribution
9 Router(config-router)# area 4 nssa no-summary
10 Router(config-router)# area 10 virtual-link 2.2.2.2
11 Router(config-router)# area 10 authentication message-digest
12
13 ! Summarization
14 Router(config-router)# area 1 range 10.1.0.0 255.255.0.0
15 Router(config-router)# area 1 range 10.2.0.0 255.255.0.0
16   advertise
17 Router(config-router)# area 1 range 10.3.0.0 255.255.0.0 not-
18   advertise
19 Router(config-router)# summary-address 192.168.0.0 255.255.0.0
```

16.3 OSPF Authentication

```
1 ! OSPF Authentication
2 ! Plain Text
3 Router(config-if)# ip ospf authentication
4 Router(config-if)# ip ospf authentication-key OSPF-PASS
5
6 ! MD5 Authentication
7 Router(config-if)# ip ospf authentication message-digest
8 Router(config-if)# ip ospf message-digest-key 1 md5 OSPF-MD5-KEY
9
10 ! Key Chain Authentication
11 Router(config)# key chain OSPF-KEYS
12 Router(config-keychain)# key 1
13 Router(config-keychain-key)# key-string OSPF-KEY-1
14 Router(config-keychain-key)# cryptographic-algorithm md5
15 Router(config-keychain-key)# accept-lifetime 00:00:00 Jan 1 2024
16   infinite
17 Router(config-keychain-key)# send-lifetime 00:00:00 Jan 1 2024
18   infinite
19 Router(config-if)# ip ospf authentication key-chain OSPF-KEYS
```

16.4 OSPF Advanced Features

```
1 ! OSPF Graceful Restart
2 Router(config)# router ospf 100
3 Router(config-router)# nsf cisco helper
4 Router(config-router)# nsf ietf helper
5 Router(config-router)# nsf ietf helper strict-lsa-checking
6
7 ! OSPF Filtering
8 Router(config-router)# distribute-list 10 in
9 Router(config-router)# distribute-list 20 out
10 Router(config-router)# area 1 filter-list prefix FILTER-1 in
11 Router(config-router)# area 1 filter-list prefix FILTER-2 out
12
13 ! OSPF Cost Manipulation
14 Router(config-if)# ip ospf cost 100
15 Router(config-if)# bandwidth 10000
16 Router(config)# auto-cost reference-bandwidth 10000
17
18 ! OSPF Database Control
19 Router(config-router)# max-lsa 50000 warning-only
20 Router(config-router)# max-metric router-lsa on-startup 300
```

16.5 OSPF Verification

```
1 ! OSPF Show Commands
2 Router# show ip ospf
3 Router# show ip ospf neighbor
4 Router# show ip ospf neighbor detail
5 Router# show ip ospf interface
6 Router# show ip ospf interface brief
7 Router# show ip ospf database
8 Router# show ip ospf database router
9 Router# show ip ospf database network
10 Router# show ip ospf database summary
11 Router# show ip ospf database asbr-summary
12 Router# show ip ospf database external
13 Router# show ip ospf database nssa-external
14 Router# show ip ospf border-routers
15 Router# show ip ospf virtual-links
16 Router# show ip ospf request-list
17 Router# show ip ospf retransmission-list
18 Router# show ip ospf events
19 Router# show ip ospf statistics
20 Router# show ip route ospf
```

16.6 OSPF Troubleshooting

```
1 ! OSPF Debug Commands
2 Router# debug ip ospf adj
3 Router# debug ip ospf events
4 Router# debug ip ospf flood
5 Router# debug ip ospf lsa-generation
6 Router# debug ip ospf packet
7 Router# debug ip ospf retransmission
8 Router# debug ip ospf spf
9 Router# debug ip ospf hello
10 Router# debug ip ospf mpls traffic-eng
11
12 ! OSPF Clear Commands
13 Router# clear ip ospf process
14 Router# clear ip ospf counters
15 Router# clear ip ospf redistribution
```

17 EIGRP

17.1 EIGRP Basics

```
1 ! EIGRP Configuration
2 Router(config)# router eigrp 100
3 Router(config-router)# eigrp router-id 1.1.1.1
4 Router(config-router)# network 10.0.0.0 0.255.255.255
5 Router(config-router)# network 192.168.1.0
6 Router(config-router)# no auto-summary
7 Router(config-router)# metric weights 0 1 0 1 0 0
8 Router(config-router)# maximum-paths 4
9 Router(config-router)# variance 2
10 Router(config-router)# traffic-share balanced
11 Router(config-router)# timers active-time 3
12 Router(config-router)# log-neighbor-changes
13 Router(config-router)# exit
14
15 ! Interface Configuration
16 Router(config-if)# ip bandwidth-percent eigrp 100 50
17 Router(config-if)# ip hello-interval eigrp 100 5
18 Router(config-if)# ip hold-time eigrp 100 15
19 Router(config-if)# ip summary-address eigrp 100 10.1.0.0
    255.255.0.0
20 Router(config-if)# ip authentication mode eigrp 100 md5
21 Router(config-if)# ip authentication key-chain eigrp 100 EIGRP-
    KEYS
```

17.2 EIGRP Named Mode

```
1 ! EIGRP Named Mode Configuration
```

```

2 Router(config)# router eigrp COMPANY
3 Router(config-router)# address-family ipv4 unicast autonomous-
  system 100
4 Router(config-router-af)# topology base
5 Router(config-router-af-topology)# exit-af-topology
6 Router(config-router-af)# network 10.0.0.0 0.255.255.255
7 Router(config-router-af)# network 192.168.1.0 0.0.0.255
8 Router(config-router-af)# af-interface GigabitEthernet0/0
9 Router(config-router-af-interface)# hello-interval 5
10 Router(config-router-af-interface)# hold-time 15
11 Router(config-router-af-interface)# authentication mode md5
12 Router(config-router-af-interface)# authentication key-chain
   EIGRP-KEYS
13 Router(config-router-af-interface)# exit-af-interface
14 Router(config-router-af)# topology base
15 Router(config-router-af-topology)# variance 2
16 Router(config-router-af-topology)# maximum-paths 4
17 Router(config-router-af-topology)# exit-af-topology
18 Router(config-router-af)# exit-address-family

```

17.3 EIGRP Authentication

```

1 ! EIGRP Authentication with Key Chain
2 Router(config)# key chain EIGRP-KEYS
3 Router(config-keychain)# key 1
4 Router(config-keychain-key)# key-string EIGRP-KEY-1
5 Router(config-keychain-key)# accept-lifetime 00:00:00 Jan 1 2024
  infinite
6 Router(config-keychain-key)# send-lifetime 00:00:00 Jan 1 2024
  infinite
7
8 ! Interface Configuration
9 Router(config-if)# ip authentication mode eigrp 100 md5
10 Router(config-if)# ip authentication key-chain eigrp 100 EIGRP-
   KEYS
11
12 ! Named Mode Authentication
13 Router(config-router-af-interface)# authentication mode md5
14 Router(config-router-af-interface)# authentication key-chain
   EIGRP-KEYS

```

17.4 EIGRP Stub Routing

```

1 ! EIGRP Stub Configuration
2 Router(config-router)# eigrp stub
3 Router(config-router)# eigrp stub connected
4 Router(config-router)# eigrp stub static
5 Router(config-router)# eigrp stub summary
6 Router(config-router)# eigrp stub redistributed

```

```
7 Router(config-router)# eigrp stub receive-only
8
9 ! Named Mode Stub
10 Router(config-router-af)# eigrp stub-site
11 Router(config-router-af)# eigrp stub connected summary leak-map
    LEAK
```

17.5 EIGRP Route Filtering

```
1 ! Route Filtering
2 Router(config-router)# distribute-list 10 in
3 Router(config-router)# distribute-list 20 out
4 Router(config-router)# distribute-list gateway GW-FILTER in
5 Router(config-router)# distribute-list prefix PREFIX-FILTER out
6
7 ! Prefix Lists
8 Router(config)# ip prefix-list EIGRP-OUT seq 10 permit
    10.1.0.0/16
9 Router(config)# ip prefix-list EIGRP-OUT seq 20 deny 0.0.0.0/0 le
    32
10
11 ! Route Maps
12 Router(config)# route-map EIGRP-FILTER permit 10
13 Router(config-route-map)# match ip address prefix-list ALLOWED
14 Router(config-route-map)# exit
```

17.6 EIGRP Verification

```
1 ! EIGRP Show Commands
2 Router# show ip eigrp neighbors
3 Router# show ip eigrp neighbors detail
4 Router# show ip eigrp topology
5 Router# show ip eigrp topology all-links
6 Router# show ip eigrp topology 10.1.1.0/24
7 Router# show ip eigrp interfaces
8 Router# show ip eigrp interfaces detail
9 Router# show ip eigrp traffic
10 Router# show ip eigrp accounting
11 Router# show ip eigrp events
12 Router# show ip eigrp plugins
13 Router# show ip route eigrp
14 Router# show ip protocols
15
16 ! EIGRP Debug Commands
17 Router# debug eigrp packets
18 Router# debug eigrp neighbors
19 Router# debug eigrp fsm
20 Router# debug eigrp notifications
```

18 BGP

18.1 BGP Basics

```
1  ! BGP Configuration
2  Router(config)# router bgp 65001
3  Router(config-router)# bgp router-id 1.1.1.1
4  Router(config-router)# bgp log-neighbor-changes
5  Router(config-router)# bgp bestpath as-path multipath-relax
6  Router(config-router)# bgp bestpath compare-routerid
7  Router(config-router)# bgp bestpath med missing-as-worst
8  Router(config-router)# bgp bestpath med confed
9  Router(config-router)# neighbor 192.168.1.2 remote-as 65002
10 Router(config-router)# neighbor 192.168.1.2 description "EBGP to
    ISP"
11 Router(config-router)# neighbor 192.168.1.2 update-source
    Loopback0
12 Router(config-router)# neighbor 192.168.1.2 ebgp-multihop 255
13 Router(config-router)# neighbor 192.168.1.2 password BGP-PASS
14 Router(config-router)# neighbor 192.168.1.2 timers 10 30
15 Router(config-router)# neighbor 192.168.1.2 advertisement-
    interval 5
16 Router(config-router)# network 10.0.0.0 mask 255.255.255.0
17 Router(config-router)# network 192.168.0.0
18 Router(config-router)# maximum-paths 4
19 Router(config-router)# maximum-paths ibgp 4
20 Router(config-router)# exit
```

18.2 iBGP Configuration

```
1  ! iBGP Full Mesh
2  Router(config)# router bgp 65001
3  Router(config-router)# neighbor 10.0.0.2 remote-as 65001
4  Router(config-router)# neighbor 10.0.0.2 update-source Loopback0
5  Router(config-router)# neighbor 10.0.0.2 next-hop-self
6  Router(config-router)# neighbor 10.0.0.3 remote-as 65001
7  Router(config-router)# neighbor 10.0.0.3 update-source Loopback0
8  Router(config-router)# neighbor 10.0.0.3 next-hop-self
9
10 ! Route Reflector
11 Router(config-router)# neighbor 10.0.0.4 remote-as 65001
12 Router(config-router)# neighbor 10.0.0.4 route-reflector-client
13 Router(config-router)# neighbor 10.0.0.5 remote-as 65001
14 Router(config-router)# neighbor 10.0.0.5 route-reflector-client
15
16 ! Confederation
17 Router(config-router)# bgp confederation identifier 65000
18 Router(config-router)# bgp confederation peers 65002 65003
```

18.3 BGP Route Filtering

```
1 ! Prefix Lists
2 Router(config)# ip prefix-list CUSTOMER-ROUTES seq 10 permit
   192.168.0.0/16 le 24
3 Router(config)# ip prefix-list CUSTOMER-ROUTES seq 20 deny
   0.0.0.0/0 le 32
4
5 ! AS Path Access Lists
6 Router(config)# ip as-path access-list 10 permit ^65001_
7 Router(config)# ip as-path access-list 10 permit ^65002$
8 Router(config)# ip as-path access-list 20 deny _65003_
9
10 ! Route Maps
11 Router(config)# route-map BGP-IN permit 10
12 Router(config-route-map)# match ip address prefix-list CUSTOMER-
   ROUTES
13 Router(config-route-map)# match as-path 10
14 Router(config-route-map)# set local-preference 200
15 Router(config-route-map)# set community 65001:100
16 Router(config-route-map)# exit
17
18 ! Apply Route Maps
19 Router(config-router)# neighbor 192.168.1.2 route-map BGP-IN in
20 Router(config-router)# neighbor 192.168.1.2 route-map BGP-OUT out
21 Router(config-router)# neighbor 192.168.1.2 prefix-list CUSTOMER-
   OUT out
22 Router(config-router)# neighbor 192.168.1.2 filter-list 10 in
```

18.4 BGP Communities

```
1 ! Community Configuration
2 Router(config)# ip bgp-community new-format
3 Router(config)# route-map SET-COMMUNITY permit 10
4 Router(config-route-map)# set community 65001:100 65001:200
   additive
5 Router(config-route-map)# set community no-export
6 Router(config-route-map)# set community no-advertise
7 Router(config-route-map)# set community internet
8 Router(config-route-map)# set community local-as
9 Router(config-route-map)# exit
10
11 ! Community Lists
12 Router(config)# ip community-list standard NO-EXPORT permit no-
   export
13 Router(config)# ip community-list expanded CUSTOMER permit
   65001:100 65001:200
14 Router(config)# ip community-list expanded CUSTOMER permit
   65001:300
15
```

```
16 ! Match Communities
17 Router(config-route-map)# match community CUSTOMER
```

18.5 BGP Route Manipulation

```
1 ! Path Selection Manipulation
2 ! 1. Weight (Cisco proprietary, local to router)
3 Router(config-router)# neighbor 192.168.1.2 weight 200
4
5 ! 2. Local Preference (within AS)
6 Router(config-route-map)# set local-preference 150
7
8 ! 3. AS Path Prepending
9 Router(config-route-map)# set as-path prepend 65001 65001 65001
10
11 ! 4. MED (between ASes)
12 Router(config-route-map)# set metric 100
13
14 ! 5. Origin Code
15 Router(config-route-map)# set origin igp
16 Router(config-route-map)# set origin egp
17 Router(config-route-map)# set origin incomplete
```

18.6 BGP Verification

```
1 ! BGP Show Commands
2 Router# show ip bgp
3 Router# show ip bgp summary
4 Router# show ip bgp neighbors
5 Router# show ip bgp neighbors 192.168.1.2
6 Router# show ip bgp neighbors 192.168.1.2 advertised-routes
7 Router# show ip bgp neighbors 192.168.1.2 received-routes
8 Router# show ip bgp neighbors 192.168.1.2 routes
9 Router# show ip bgp community
10 Router# show ip bgp community 65001:100
11 Router# show ip bgp dampened-paths
12 Router# show ip bgp flap-statistics
13 Router# show ip bgp regexp ^65001
14 Router# show ip bgp ipv4 unicast
15 Router# show ip bgp vpnv4 vrf CUSTOMER
16 Router# show bgp all summary
17 Router# show bgp l2vpn evpn
18
19 ! BGP Debug Commands
20 Router# debug ip bgp
21 Router# debug ip bgp updates
22 Router# debug ip bgp keepalives
23 Router# debug ip bgp dampening
24 Router# debug ip bgp events
```

```
25 Router# debug ip bgp filters
```

19 Access Control Lists

19.1 Standard ACLs

```
1 ! Standard Numbered ACLs (1-99, 1300-1999)
2 Router(config)# access-list 10 permit 192.168.1.0 0.0.0.255
3 Router(config)# access-list 10 permit 10.0.0.0 0.255.255.255
4 Router(config)# access-list 10 deny any log
5
6 ! Standard Named ACLs
7 Router(config)# ip access-list standard MANAGEMENT
8 Router(config-std-nacl)# permit 192.168.100.0 0.0.0.255
9 Router(config-std-nacl)# permit 10.0.0.0 0.255.255.255
10 Router(config-std-nacl)# deny any log
11 Router(config-std-nacl)# exit
12
13 ! Apply to Interface
14 Router(config-if)# ip access-group 10 in
15 Router(config-if)# ip access-group MANAGEMENT out
```

19.2 Extended ACLs

```
1 ! Extended Numbered ACLs (100-199, 2000-2699)
2 Router(config)# access-list 100 permit tcp 192.168.1.0 0.0.0.255
   any eq 80
3 Router(config)# access-list 100 permit tcp 192.168.1.0 0.0.0.255
   any eq 443
4 Router(config)# access-list 100 permit udp 192.168.1.0 0.0.0.255
   any eq 53
5 Router(config)# access-list 100 permit icmp 192.168.1.0 0.0.0.255
   any echo
6 Router(config)# access-list 100 permit icmp 192.168.1.0 0.0.0.255
   any echo-reply
7 Router(config)# access-list 100 deny ip any any log
8
9 ! Extended Named ACLs
10 Router(config)# ip access-list extended INTERNET-ACCESS
11 Router(config-ext-nacl)# permit tcp 192.168.1.0 0.0.0.255 any eq
   www
12 Router(config-ext-nacl)# permit tcp 192.168.1.0 0.0.0.255 any eq
   443
13 Router(config-ext-nacl)# permit udp 192.168.1.0 0.0.0.255 any eq
   domain
14 Router(config-ext-nacl)# permit icmp 192.168.1.0 0.0.0.255 any
15 Router(config-ext-nacl)# deny ip any any log-input
16 Router(config-ext-nacl)# exit
```

```
17
18 ! Apply to Interface
19 Router(config-if)# ip access-group INTERNET-ACCESS out
```

19.3 Advanced ACL Features

```
1 ! Time-Based ACLs
2 Router(config)# time-range WORK-HOURS
3 Router(config-time-range)# periodic weekdays 9:00 to 17:00
4 Router(config-time-range)# absolute start 00:00 1 Jan 2024 end
   23:59 31 Dec 2024
5
6 Router(config)# ip access-list extended WORK-ACL
7 Router(config-ext-nacl)# permit tcp any any eq 80 time-range WORK
   -HOURS
8 Router(config-ext-nacl)# deny ip any any
9 Router(config-ext-nacl)# exit
10
11 ! Reflexive ACLs
12 Router(config)# ip access-list extended OUTBOUND
13 Router(config-ext-nacl)# permit tcp any any reflect TCP-TRAFFIC
14 Router(config-ext-nacl)# permit udp any any reflect UDP-TRAFFIC
15 Router(config-ext-nacl)# permit icmp any any reflect ICMP-TRAFFIC
16 Router(config-ext-nacl)# exit
17
18 Router(config)# ip access-list extended INBOUND
19 Router(config-ext-nacl)# evaluate TCP-TRAFFIC
20 Router(config-ext-nacl)# evaluate UDP-TRAFFIC
21 Router(config-ext-nacl)# evaluate ICMP-TRAFFIC
22 Router(config-ext-nacl)# deny ip any any log
23 Router(config-ext-nacl)# exit
24
25 ! Dynamic/Lock-and-Key ACLs
26 Router(config)# username remote-user password RemotePass123
27 Router(config)# access-list 110 dynamic REMOTE-ACCESS timeout 120
   permit ip any any
28 Router(config)# line vty 0 4
29 Router(config-line)# login local
30 Router(config-line)# autocmd command access-enable host timeout 10
```

19.4 ACL Optimization & Management

```
1 ! ACL Sequence Numbers
2 Router(config)# ip access-list extended OPTIMIZED-ACL
3 Router(config-ext-nacl)# 10 permit tcp any any eq 80
4 Router(config-ext-nacl)# 20 permit tcp any any eq 443
5 Router(config-ext-nacl)# 30 deny tcp any any eq 3389
6 Router(config-ext-nacl)# 40 permit ip any any
7 Router(config-ext-nacl)# exit
```

```

8
9 ! Insert/Delete/Resequence
10 Router(config-ext-nacl)# 15 permit tcp any any eq 22
11 Router(config-ext-nacl)# no 20
12 Router(config-ext-nacl)# resequence 10 10 10
13
14 ! ACL Logging
15 Router(config-ext-nacl)# permit tcp any any eq 80 log
16 Router(config-ext-nacl)# permit tcp any any eq 80 log-input
17 Router(config-ext-nacl)# deny ip any any log
18
19 ! ACL Statistics
20 Router# show access-lists
21 Router# show access-lists INTERNET-ACCESS
22 Router# show ip access-list
23 Router# clear access-list counters
24 Router# clear access-list counters INTERNET-ACCESS

```

19.5 Infrastructure ACLs (iACLs)

```

1 ! iACL for Infrastructure Protection
2 Router(config)# ip access-list extended INFRASTRUCTURE-ACL
3 ! Permit BGP
4 Router(config-ext-nacl)# permit tcp host 192.168.1.2 eq bgp host
   192.168.1.1
5 Router(config-ext-nacl)# permit tcp host 192.168.1.2 host
   192.168.1.1 eq bgp
6 ! Permit OSPF
7 Router(config-ext-nacl)# permit ospf any any
8 ! Permit EIGRP
9 Router(config-ext-nacl)# permit eigrp any any
10 ! Permit ICMP for Management
11 Router(config-ext-nacl)# permit icmp 192.168.100.0 0.0.0.255 any
   echo
12 Router(config-ext-nacl)# permit icmp 192.168.100.0 0.0.0.255 any
   echo-reply
13 Router(config-ext-nacl)# permit icmp 192.168.100.0 0.0.0.255 any
   ttl-exceeded
14 Router(config-ext-nacl)# permit icmp 192.168.100.0 0.0.0.255 any
   unreachable
15 ! Permit SSH/HTTPS for Management
16 Router(config-ext-nacl)# permit tcp 192.168.100.0 0.0.0.255 any
   eq 22
17 Router(config-ext-nacl)# permit tcp 192.168.100.0 0.0.0.255 any
   eq 443
18 ! Permit NTP
19 Router(config-ext-nacl)# permit udp 192.168.100.0 0.0.0.255 any
   eq 123
20 ! Permit Syslog

```

```

21 Router(config-ext-nacl)# permit udp 192.168.100.0 0.0.0.255 any
    eq 514
22 ! Deny and Log Everything Else
23 Router(config-ext-nacl)# deny ip any any log-input
24 Router(config-ext-nacl)# exit
25
26 ! Apply iACL
27 Router(config-if)# ip access-group INFRASTRUCTURE-ACL in

```

20 Zone-Based Firewall

20.1 ZBFW Configuration

```

1 ! Zone Configuration
2 Router(config)# zone security INSIDE
3 Router(config-sec-zone)# description "Trusted Internal Network"
4 Router(config-sec-zone)# exit
5
6 Router(config)# zone security OUTSIDE
7 Router(config-sec-zone)# description "Untrusted External Network"
8 Router(config-sec-zone)# exit
9
10 Router(config)# zone security DMZ
11 Router(config-sec-zone)# description "Demilitarized Zone"
12 Router(config-sec-zone)# exit
13
14 ! Assign Interfaces to Zones
15 Router(config)# interface GigabitEthernet0/0
16 Router(config-if)# zone-member security INSIDE
17 Router(config-if)# exit
18
19 Router(config)# interface GigabitEthernet0/1
20 Router(config-if)# zone-member security OUTSIDE
21 Router(config-if)# exit
22
23 Router(config)# interface GigabitEthernet0/2
24 Router(config-if)# zone-member security DMZ
25 Router(config-if)# exit

```

20.2 Class Maps & Policy Maps

```

1 ! Class Maps for Traffic Classification
2 Router(config)# class-map type inspect match-any INSIDE-OUTSIDE-
    CMAP
3 Router(config-cmap)# match protocol http
4 Router(config-cmap)# match protocol https
5 Router(config-cmap)# match protocol dns
6 Router(config-cmap)# match protocol icmp

```

```

7 Router(config-cmap)# exit
8
9 Router(config)# class-map type inspect match-any OUTSIDE-DMZ-CMAP
10 Router(config-cmap)# match protocol tcp
11 Router(config-cmap)# match access-group name DMZ-SERVERS
12 Router(config-cmap)# exit
13
14 ! Policy Maps for Actions
15 Router(config)# policy-map type inspect INSIDE-OUTSIDE-PMAP
16 Router(config-pmap)# class type inspect INSIDE-OUTSIDE-CMAP
17 Router(config-pmap-c)# inspect
18 Router(config-pmap-c)# pass
19 Router(config-pmap-c)# drop log
20 Router(config-pmap-c)# exit
21 Router(config-pmap)# class class-default
22 Router(config-pmap-c)# drop log
23 Router(config-pmap-c)# exit
24 Router(config-pmap)# exit
25
26 Router(config)# policy-map type inspect OUTSIDE-DMZ-PMAP
27 Router(config-pmap)# class type inspect OUTSIDE-DMZ-CMAP
28 Router(config-pmap-c)# inspect
29 Router(config-pmap-c)# exit
30 Router(config-pmap)# class class-default
31 Router(config-pmap-c)# drop log
32 Router(config-pmap-c)# exit
33 Router(config-pmap)# exit

```

20.3 Zone Pairs & Service Policies

```

1 ! Zone Pairs
2 Router(config)# zone-pair security INSIDE-TO-OUTSIDE source
   INSIDE destination OUTSIDE
3 Router(config-sec-zone-pair)# description "Internal to External
   Traffic"
4 Router(config-sec-zone-pair)# service-policy type inspect INSIDE-
   OUTSIDE-PMAP
5 Router(config-sec-zone-pair)# exit
6
7 Router(config)# zone-pair security OUTSIDE-TO-DMZ source OUTSIDE
   destination DMZ
8 Router(config-sec-zone-pair)# description "External to DMZ
   Traffic"
9 Router(config-sec-zone-pair)# service-policy type inspect OUTSIDE-
   -DMZ-PMAP
10 Router(config-sec-zone-pair)# exit
11
12 Router(config)# zone-pair security DMZ-TO-OUTSIDE source DMZ
   destination OUTSIDE

```

```

13 Router(config-sec-zone-pair)# description "DMZ to External
    Traffic"
14 Router(config-sec-zone-pair)# service-policy type inspect DMZ-
    OUTSIDE-PMAP
15 Router(config-sec-zone-pair)# exit
16
17 ! Default Deny (Self Zone)
18 Router(config)# zone-pair security SELF-TO-ANY source self
    destination any
19 Router(config-sec-zone-pair)# service-policy type inspect SELF-
    PMAP
20 Router(config-sec-zone-pair)# exit
21
22 Router(config)# zone-pair security ANY-TO-SELF source any
    destination self
23 Router(config-sec-zone-pair)# service-policy type inspect SELF-
    PMAP
24 Router(config-sec-zone-pair)# exit

```

20.4 ZBFW Parameters & Tuning

```

1 ! Timeouts and Thresholds
2 Router(config)# parameter-map type inspect global
3 Router(config-profile)# alert on
4 Router(config-profile)# audit-trail on
5 Router(config-profile)# session total 200000
6 Router(config-profile)# tcp synwait-time 30
7 Router(config-profile)# tcp finwait-time 5
8 Router(config-profile)# tcp idle-time 3600
9 Router(config-profile)# udp idle-time 30
10 Router(config-profile)# dns-timeout 5
11 Router(config-profile)# max-incomplete high 800
12 Router(config-profile)# max-incomplete low 400
13 Router(config-profile)# one-minute high 800
14 Router(config-profile)# one-minute low 400
15 Router(config-profile)# tcp max-incomplete host 50 block-time 0
16 Router(config-profile)# exit
17
18 ! Protocol-specific Parameters
19 Router(config)# parameter-map type inspect http
20 Router(config-profile)# protocol-violation action drop-connection
    log
21 Router(config-profile)# body-match-max-size 1024
22 Router(config-profile)# request-method rfc get head post put
    delete options trace connect
23 Router(config-profile)# exit

```

20.5 ZBFW Verification

```

1  ! ZBFW Show Commands
2  Router# show zone security
3  Router# show zone-pair security
4  Router# show policy-map type inspect zone-pair sessions
5  Router# show policy-map type inspect zone-pair stats
6  Router# show policy-map type inspect zone-pair INSIDE-TO-OUTSIDE
7  Router# show zone-pair security INSIDE-TO-OUTSIDE
8  Router# show policy-map type inspect all
9  Router# show class-map type inspect
10 Router# show parameter-map type inspect
11 Router# show policy-firewall stats
12
13 ! ZBFW Debug Commands
14 Router# debug policy-firewall
15 Router# debug policy-firewall events
16 Router# debug policy-firewall protocol
17 Router# debug zone-pair
18 Router# debug zone security
19
20 ! Clear Commands
21 Router# clear policy-firewall stats
22 Router# clear policy-firewall sessions
23 Router# clear zone-pair counter

```

21 IPsec VPN

21.1 Site-to-Site IPsec VPN

```

1  ! Phase 1: IKE Policy
2  Router(config)# crypto isakmp policy 10
3  Router(config-isakmp)# encryption aes 256
4  Router(config-isakmp)# hash sha256
5  Router(config-isakmp)# authentication pre-share
6  Router(config-isakmp)# group 14
7  Router(config-isakmp)# lifetime 86400
8  Router(config-isakmp)# exit
9
10 Router(config)# crypto isakmp key VPN-KEY-123 address 203.0.113.2
11 Router(config)# crypto isakmp keepalive 10 periodic
12
13 ! Phase 2: IPsec Transform Set
14 Router(config)# crypto ipsec transform-set TS-ESP-AES256-SHA256
    esp-aes 256 esp-sha256-hmac
15 Router(config-crypto-trans)# mode tunnel
16 Router(config-crypto-trans)# exit
17
18 Router(config)# crypto ipsec transform-set TS-ESP-AES128-SHA esp-
    aes esp-sha-hmac
19 Router(config-crypto-trans)# mode tunnel

```

```

20 Router(config-crypto-trans)# exit
21
22 ! IPSec Profile (New Style)
23 Router(config)# crypto ipsec profile IPSEC-PROFILE
24 Router(config-ipsec-profile)# set transform-set TS-ESP-AES256-
    SHA256
25 Router(config-ipsec-profile)# set pfs group14
26 Router(config-ipsec-profile)# set security-association lifetime
    seconds 28800
27 Router(config-ipsec-profile)# set security-association lifetime
    kilobytes 4608000
28 Router(config-ipsec-profile)# exit

```

21.2 Crypto Map Configuration

```

1 ! Crypto Map (Traditional)
2 Router(config)# crypto map CRYPTO-MAP 10 ipsec-isakmp
3 Router(config-crypto-map)# description "VPN to Branch Office"
4 Router(config-crypto-map)# set peer 203.0.113.2
5 Router(config-crypto-map)# set transform-set TS-ESP-AES256-SHA256
6 Router(config-crypto-map)# set pfs group14
7 Router(config-crypto-map)# match address VPN-INTERESTING-TRAFFIC
8 Router(config-crypto-map)# set security-association lifetime
    seconds 28800
9 Router(config-crypto-map)# set security-association lifetime
    kilobytes 4608000
10 Router(config-crypto-map)# exit
11
12 ! Interesting Traffic ACL
13 Router(config)# ip access-list extended VPN-INTERESTING-TRAFFIC
14 Router(config-ext-nacl)# permit ip 192.168.1.0 0.0.0.255
    192.168.2.0 0.0.0.255
15 Router(config-ext-nacl)# permit ip 192.168.1.0 0.0.0.255
    192.168.3.0 0.0.0.255
16 Router(config-ext-nacl)# exit
17
18 ! Apply Crypto Map to Interface
19 Router(config)# interface GigabitEthernet0/0
20 Router(config-if)# crypto map CRYPTO-MAP
21 Router(config-if)# exit

```

21.3 VTI (Virtual Tunnel Interface)

```

1 ! VTI Configuration
2 Router(config)# interface Tunnel0
3 Router(config-if)# description "VTI to Branch Office"
4 Router(config-if)# ip address 10.255.255.1 255.255.255.252
5 Router(config-if)# tunnel source GigabitEthernet0/0
6 Router(config-if)# tunnel destination 203.0.113.2

```

```

7 Router(config-if)# tunnel mode ipsec ipv4
8 Router(config-if)# tunnel protection ipsec profile IPSEC-PROFILE
9 Router(config-if)# exit
10
11 ! Static Route over VTI
12 Router(config)# ip route 192.168.2.0 255.255.255.0 Tunnel0
13 Router(config)# ip route 192.168.3.0 255.255.255.0 Tunnel0

```

21.4 DMVPN Configuration

```

1 ! DMVPN Phase 3
2 Router(config)# interface Tunnel100
3 Router(config-if)# description "DMVPN Hub"
4 Router(config-if)# ip address 10.255.255.1 255.255.255.0
5 Router(config-if)# ip mtu 1400
6 Router(config-if)# ip nhrp authentication DMVPN-KEY
7 Router(config-if)# ip nhrp map multicast dynamic
8 Router(config-if)# ip nhrp network-id 100
9 Router(config-if)# ip nhrp redirect
10 Router(config-if)# ip tcp adjust-mss 1360
11 Router(config-if)# tunnel source GigabitEthernet0/0
12 Router(config-if)# tunnel mode gre multipoint
13 Router(config-if)# tunnel key 100
14 Router(config-if)# tunnel protection ipsec profile IPSEC-PROFILE
15 Router(config-if)# exit
16
17 ! DMVPN Spoke
18 Router(config)# interface Tunnel100
19 Router(config-if)# description "DMVPN Spoke"
20 Router(config-if)# ip address 10.255.255.2 255.255.255.0
21 Router(config-if)# ip mtu 1400
22 Router(config-if)# ip nhrp authentication DMVPN-KEY
23 Router(config-if)# ip nhrp map 10.255.255.1 203.0.113.1
24 Router(config-if)# ip nhrp map multicast 203.0.113.1
25 Router(config-if)# ip nhrp network-id 100
26 Router(config-if)# ip nhrp nhs 10.255.255.1
27 Router(config-if)# ip nhrp registration timeout 300
28 Router(config-if)# ip nhrp shortcut
29 Router(config-if)# ip tcp adjust-mss 1360
30 Router(config-if)# tunnel source GigabitEthernet0/0
31 Router(config-if)# tunnel mode gre multipoint
32 Router(config-if)# tunnel key 100
33 Router(config-if)# tunnel protection ipsec profile IPSEC-PROFILE
34 Router(config-if)# exit

```

21.5 GETVPN Configuration

```

1 ! GETVPN Key Server
2 Router(config)# crypto gdoi group GETVPN-GROUP

```

```

3 Router(config-gdoi-group)# identity number 100
4 Router(config-gdoi-group)# server local
5 Router(config-gdoi-group-server)# rekey lifetime seconds 86400
6 Router(config-gdoi-group-server)# rekey retransmit 10 number 3
7 Router(config-gdoi-group-server)# rekey authentication mypubkey
  rsa GETVPN-KEYS
8 Router(config-gdoi-group-server)# rekey transport unicast
9 Router(config-gdoi-group-server)# sa ipsec 1
10 Router(config-gdoi-group-server-sa)# profile IPSEC-PROFILE
11 Router(config-gdoi-group-server-sa)# match address ipv4 GETVPN-
  ACL
12 Router(config-gdoi-group-server-sa)# replay counter window-size
  64
13 Router(config-gdoi-group-server-sa)# exit
14 Router(config-gdoi-group-server)# address ipv4 192.168.100.10
15 Router(config-gdoi-group-server)# exit
16 Router(config-gdoi-group)# exit
17
18 ! GETVPN Group Member
19 Router(config)# crypto gdoi group GETVPN-GROUP
20 Router(config-gdoi-group)# identity number 100
21 Router(config-gdoi-group)# server address ipv4 192.168.100.10
22 Router(config-gdoi-group)# exit
23
24 Router(config)# interface Tunnel0
25 Router(config-if)# tunnel mode ipsec ipv4
26 Router(config-if)# tunnel protection gdoi profile GETVPN-GROUP

```

21.6 IPSec Verification

```

1 ! IPSec Show Commands
2 Router# show crypto isakmp sa
3 Router# show crypto isakmp peers
4 Router# show crypto ipsec sa
5 Router# show crypto ipsec sa detail
6 Router# show crypto ipsec sa interface GigabitEthernet0/0
7 Router# show crypto session
8 Router# show crypto session detail
9 Router# show crypto engine connections active
10 Router# show crypto map
11 Router# show crypto map interface GigabitEthernet0/0
12 Router# show crypto gdoi
13 Router# show crypto gdoi ks
14 Router# show crypto gdoi gm
15 Router# show dmvpn
16 Router# show dmvpn detail
17 Router# show nhrp
18 Router# show nhrp traffic
19
20 ! IPSec Debug Commands

```

```

21 Router# debug crypto isakmp
22 Router# debug crypto ipsec
23 Router# debug crypto engine
24 Router# debug crypto gdoi
25 Router# debug nhrp
26 Router# debug tunnel
27
28 ! Clear Commands
29 Router# clear crypto isakmp
30 Router# clear crypto sa
31 Router# clear crypto session
32 Router# clear crypto gdoi

```

22 VoIP Configuration

22.1 VoIP Basics

```

1 ! DHCP for VoIP Phones
2 Router(config)# ip dhcp pool VOICE-POOL
3 Router(dhcp-config)# network 10.10.0.0 255.255.255.0
4 Router(dhcp-config)# default-router 10.10.0.1
5 Router(dhcp-config)# option 150 ip 10.0.0.10 ! Call Manager
6 Router(dhcp-config)# option 66 ip 10.0.0.10 ! TFTP Server
7 Router(dhcp-config)# lease 8
8 Router(dhcp-config)# exit
9
10 ! Voice VLAN Configuration
11 Switch(config)# vlan 110
12 Switch(config-vlan)# name VOICE
13 Switch(config-vlan)# exit
14
15 Switch(config)# interface gigabitEthernet 1/0/1
16 Switch(config-if)# switchport voice vlan 110
17 Switch(config-if)# switchport priority extend cos 0
18 Switch(config-if)# auto qos voip trust
19 Switch(config-if)# spanning-tree portfast
20 Switch(config-if)# spanning-tree bpduguard enable

```

22.2 Call Manager Express

```

1 ! CME Basic Configuration
2 Router(config)# telephony-service
3 Router(config-telephony)# max-ephones 24
4 Router(config-telephony)# max-dn 48
5 Router(config-telephony)# ip source-address 10.0.0.1 port 2000
6 Router(config-telephony)# auto assign 1 to 24
7 Router(config-telephony)# system message "Company Phone System"
8 Router(config-telephony)# voicemail 5000

```

```

9 Router(config-telephony)# max-conferences 8 gain -6
10 Router(config-telephony)# transfer-system full-consult
11 Router(config-telephony)# exit
12
13 ! Ephone Configuration
14 Router(config)# ephone-dn 1 dual-line
15 Router(config-ephone-dn)# number 1001
16 Router(config-ephone-dn)# name "John Doe"
17 Router(config-ephone-dn)# exit
18
19 Router(config)# ephone 1
20 Router(config-ephone)# device-security-mode
21 Router(config-ephone)# mac-address 0050.7966.6800
22 Router(config-ephone)# type 7960
23 Router(config-ephone)# button 1:1
24 Router(config-ephone)# exit

```

22.3 Dial Peers

```

1 ! POTS Dial Peers
2 Router(config)# dial-peer voice 1 pots
3 Router(config-dial-peer)# destination-pattern 9T
4 Router(config-dial-peer)# port 0/0/0:23
5 Router(config-dial-peer)# forward-digits all
6 Router(config-dial-peer)# exit
7
8 ! VoIP Dial Peers
9 Router(config)# dial-peer voice 100 voip
10 Router(config-dial-peer)# destination-pattern 2...
11 Router(config-dial-peer)# session target ipv4:10.0.0.2
12 Router(config-dial-peer)# codec g711ulaw
13 Router(config-dial-peer)# dtmf-relay h245-alphanumeric
14 Router(config-dial-peer)# fax rate disable
15 Router(config-dial-peer)# exit
16
17 ! SIP Trunk Configuration
18 Router(config)# voice service voip
19 Router(config-voi-serv)# allow-connections sip to sip
20 Router(config-voi-serv)# sip
21 Router(config-voi-serv-sip)# bind control source-interface
    Loopback0
22 Router(config-voi-serv-sip)# bind media source-interface
    Loopback0
23 Router(config-voi-serv-sip)# registrar server expires max 3600
    min 3600
24 Router(config-voi-serv-sip)# exit

```

23 QoS Implementation

23.1 QoS Classification & Marking

```
1  ! Class Maps
2  Router(config)# class-map match-any VOICE
3  Router(config-cmap)# match dscp ef
4  Router(config-cmap)# match ip rtp 16384 16383
5  Router(config-cmap)# exit
6
7  Router(config)# class-map match-any VIDEO
8  Router(config-cmap)# match dscp af41 af42 af43
9  Router(config-cmap)# match ip precedence 4
10 Router(config-cmap)# exit
11
12 Router(config)# class-map match-any CALL-SIGNALING
13 Router(config-cmap)# match dscp cs3
14 Router(config-cmap)# match dscp af31
15 Router(config-cmap)# exit
16
17 Router(config)# class-map match-any INTERACTIVE-VIDEO
18 Router(config-cmap)# match dscp af21 af22 af23
19 Router(config-cmap)# exit
20
21 Router(config)# class-map match-any NETWORK-CONTROL
22 Router(config-cmap)# match dscp cs6
23 Router(config-cmap)# exit
24
25 Router(config)# class-map match-any BULK-DATA
26 Router(config-cmap)# match dscp af11 af12 af13
27 Router(config-cmap)# exit
28
29 Router(config)# class-map match-any SCAVENGER
30 Router(config-cmap)# match dscp cs1
31 Router(config-cmap)# exit
```

23.2 Policy Maps

```
1  ! LLQ for Voice and Video
2  Router(config)# policy-map WAN-EDGE-POLICY
3  Router(config-pmap)# class VOICE
4  Router(config-pmap-c)# priority percent 10
5  Router(config-pmap-c)# exit
6  Router(config-pmap)# class VIDEO
7  Router(config-pmap-c)# priority percent 23
8  Router(config-pmap-c)# exit
9  Router(config-pmap)# class CALL-SIGNALING
10 Router(config-pmap-c)# bandwidth percent 5
11 Router(config-pmap-c)# exit
```

```

12 Router(config-pmap)# class INTERACTIVE-VIDEO
13 Router(config-pmap-c)# bandwidth percent 10
14 Router(config-pmap-c)# random-detect
15 Router(config-pmap-c)# exit
16 Router(config-pmap)# class NETWORK-CONTROL
17 Router(config-pmap-c)# bandwidth percent 5
18 Router(config-pmap-c)# exit
19 Router(config-pmap)# class class-default
20 Router(config-pmap-c)# bandwidth remaining percent 25
21 Router(config-pmap-c)# random-detect
22 Router(config-pmap-c)# exit
23 Router(config-pmap)# exit

```

23.3 AutoQoS

```

1 ! AutoQoS VoIP
2 Router(config)# interface GigabitEthernet0/0
3 Router(config-if)# auto qos voip trust
4 Router(config-if)# service-policy output AUTOQOS-POLICY
5 Router(config-if)# exit
6
7 ! AutoQoS for Enterprise
8 Router(config-if)# auto qos trust dscp
9 Router(config-if)# auto discovery qos
10
11 ! Verification
12 Router# show auto qos
13 Router# show auto discovery qos
14 Router# show policy-map interface GigabitEthernet0/0

```

23.4 QoS Verification

```

1 ! QoS Show Commands
2 Router# show policy-map
3 Router# show policy-map interface
4 Router# show policy-map interface GigabitEthernet0/0 input
5 Router# show policy-map interface GigabitEthernet0/0 output
6 Router# show class-map
7 Router# show mls qos
8 Router# show mls qos interface
9 Router# show mls qos maps
10 Router# show mls qos statistics
11 Router# show mls qos queueing
12 Router# show queueing interface GigabitEthernet0/0
13
14 ! QoS Debug Commands
15 Router# debug qos
16 Router# debug policy-firewall
17 Router# debug nbar

```

24 Wireless Networking

24.1 WLC Configuration

```
1 ! WLC Basic Setup
2 WLC(config)# config interface create MANAGEMENT 99 192.168.99.10
   255.255.255.0 192.168.99.1
3 WLC(config)# config interface create GUEST 100 192.168.100.10
   255.255.255.0 192.168.100.1
4 WLC(config)# config interface create CORPORATE 10 192.168.10.10
   255.255.255.0 192.168.10.1
5
6 ! Management User
7 WLC(config)# config mgmtuser add admin AdminPass123! read-write
8 WLC(config)# config mgmtuser add operator Oper@tor123 read-only
9
10 ! WLAN Configuration
11 WLC(config)# config wlan create 1 CORPORATE-WIFI
12 WLC(config-wlan)# config wlan ssid 1 CORPORATE-WIFI
13 WLC(config-wlan)# config wlan interface 1 CORPORATE
14 WLC(config-wlan)# config wlan security wpa akm psk set-key ascii
   WlF1-Passw0rd 1
15 WLC(config-wlan)# config wlan security wpa akm psk enable 1
16 WLC(config-wlan)# config wlan enable 1
17 WLC(config-wlan)# exit
18
19 ! Guest WLAN
20 WLC(config)# config wlan create 2 GUEST-WIFI
21 WLC(config-wlan)# config wlan ssid 2 GUEST-WIFI
22 WLC(config-wlan)# config wlan interface 2 GUEST
23 WLC(config-wlan)# config wlan security web-auth enable 2
24 WLC(config-wlan)# config wlan security web-auth parameter-map
   GUEST-PARAMS 2
25 WLC(config-wlan)# config wlan enable 2
26 WLC(config-wlan)# exit
```

24.2 AP Management

```
1 ! AP Configuration
2 WLC(config)# config ap mode local AP01
3 WLC(config)# config ap primary-base WLC01 AP01 192.168.99.10
4 WLC(config)# config ap country US AP01
5 WLC(config)# config ap group-name FLOOR1 AP01
6 WLC(config)# config ap name AP01 0050.7966.6800
7 WLC(config)# config ap tx-power 1 AP01
8 WLC(config)# config ap led-state enable AP01
9
10 ! AP Groups
11 WLC(config)# config wlan apgroup add-profile AP-GROUP
```

```
12 WLC(config)# config wlan apgroup add-profile AP-GROUP-FLOOR1
13 WLC(config-apgroup)# config wlan apgroup interface-mapping add AP
   -GROUP-FLOOR1 1 CORPORATE
14 WLC(config-apgroup)# config wlan apgroup interface-mapping add AP
   -GROUP-FLOOR1 2 GUEST
15 WLC(config-apgroup)# config ap group-name AP-GROUP-FLOOR1 AP01
```

24.3 Wireless Security

```
1 ! WPA2-Enterprise
2 WLC(config)# config wlan security wpa akm 802.1x enable 1
3 WLC(config)# config wlan security wpa wpa2 enable 1
4 WLC(config)# config wlan security dot1x authentication-server 1
   192.168.100.20
5 WLC(config)# config wlan security dot1x accounting-server 1
   192.168.100.20
6
7 ! WPA3 Configuration
8 WLC(config)# config wlan security wpa akm sae enable 1
9 WLC(config)# config wlan security wpa wpa3 enable 1
10 WLC(config)# config wlan security wpa akm sae psk set-key ascii
   WPA3-Passw0rd 1
11
12 ! MAC Filtering
13 WLC(config)# config macfilter add 0050.7966.6800 "CEO iPhone"
14 WLC(config)# config wlan macfilter 1 enable
```

24.4 Wireless Troubleshooting

```
1 ! WLC Show Commands
2 WLC# show ap summary
3 WLC# show ap config general AP01
4 WLC# show wlan summary
5 WLC# show wlan 1
6 WLC# show client summary
7 WLC# show client detail 0050.7966.6800
8 WLC# show interface summary
9 WLC# show interface detail MANAGEMENT
10 WLC# show mobility summary
11 WLC# show mobility anchor
12 WLC# show radius summary
13 WLC# show mesh summary
14 WLC# show cdp neighbors
15 WLC# show log
16
17 ! Debug Commands
18 WLC# debug client 0050.7966.6800
19 WLC# debug dot1x events enable
20 WLC# debug wps events enable
```

```
21 WLC# debug mobility handoff enable
22 WLC# debug aaa events enable
```

25 Data Center Technologies

25.1 Nexus Switching

```
1 ! Nexus Basic Configuration
2 Nexus(config)# feature interface-vlan
3 Nexus(config)# feature lacp
4 Nexus(config)# feature vpc
5 Nexus(config)# feature ospf
6 Nexus(config)# feature bgp
7 Nexus(config)# feature ssh
8
9 ! VPC Configuration
10 Nexus(config)# vpc domain 1
11 Nexus(config-vpc-domain)# peer-keepalive destination 10.0.0.2
    source 10.0.0.1
12 Nexus(config-vpc-domain)# peer-gateway
13 Nexus(config-vpc-domain)# auto-recovery
14 Nexus(config-vpc-domain)# ip arp synchronize
15 Nexus(config-vpc-domain)# delay restore 300
16 Nexus(config-vpc-domain)# exit
17
18 Nexus(config)# interface port-channel 10
19 Nexus(config-if)# vpc 10
20 Nexus(config-if)# switchport mode trunk
21 Nexus(config-if)# switchport trunk allowed vlan 10,20,30
22 Nexus(config-if)# spanning-tree port type network
23 Nexus(config-if)# exit
```

25.2 FCoE Configuration

```
1 ! FCoE Setup
2 Nexus(config)# feature fcoe
3 Nexus(config)# vsan database
4 Nexus(config-vsan-db)# vsan 100
5 Nexus(config-vsan-db)# exit
6
7 Nexus(config)# interface fc1/1
8 Nexus(config-if)# switchport mode f
9 Nexus(config-if)# switchport trunk allowed vsan 100
10 Nexus(config-if)# no shutdown
11 Nexus(config-if)# exit
12
13 Nexus(config)# fcoe vsan 100
```

26.2 MPLS L3VPN

```
1 ! VRF Configuration
2 Router(config)# ip vrf CUSTOMER-A
3 Router(config-vrf)# rd 65001:100
4 Router(config-vrf)# route-target export 65001:100
5 Router(config-vrf)# route-target import 65001:100
6 Router(config-vrf)# exit
7
8 ! VRF Interface
9 Router(config)# interface GigabitEthernet0/0.100
10 Router(config-if)# encapsulation dot1Q 100
11 Router(config-if)# ip vrf forwarding CUSTOMER-A
12 Router(config-if)# ip address 192.168.1.1 255.255.255.0
13 Router(config-if)# exit
14
15 ! MP-BGP for VPNv4
16 Router(config)# router bgp 65001
17 Router(config-router)# neighbor 2.2.2.2 remote-as 65001
18 Router(config-router)# neighbor 2.2.2.2 update-source Loopback0
19 Router(config-router)# address-family vpnv4
20 Router(config-router-af)# neighbor 2.2.2.2 activate
21 Router(config-router-af)# neighbor 2.2.2.2 send-community
    extended
22 Router(config-router-af)# exit-address-family
23 Router(config-router)# address-family ipv4 vrf CUSTOMER-A
24 Router(config-router-af)# redistribute connected
25 Router(config-router-af)# redistribute static
26 Router(config-router-af)# exit-address-family
```

27 Systematic Troubleshooting

27.1 OSI Model Troubleshooting

```
1 ! Layer 1 - Physical
2 Router# show interfaces status
3 Router# show interfaces description
4 Router# show controllers
5 Router# show environment
6 Router# show power
7 Router# show module
8
9 ! Layer 2 - Data Link
10 Router# show interfaces
11 Router# show interfaces trunk
12 Router# show spanning-tree
13 Router# show vlan
14 Router# show mac address-table
15 Router# show cdp neighbors
```

```
16 Router# show lldp neighbors
17
18 ! Layer 3 - Network
19 Router# show ip interface brief
20 Router# show ip route
21 Router# show ip protocols
22 Router# show ip ospf neighbor
23 Router# show ip eigrp neighbors
24 Router# show ip bgp summary
25 Router# show arp
26
27 ! Layer 4 - Transport
28 Router# show tcp brief
29 Router# show tcp statistics
30 Router# show udp
31
32 ! Layer 7 - Application
33 Router# show logging
34 Router# show processes cpu
35 Router# show memory
36 Router# show buffers
```

27.2 Troubleshooting Methodology

```
1 ! 1. Define the Problem
2 ! 2. Gather Information
3 Router# show tech-support
4 Router# show logging
5 Router# show clock
6 Router# show version
7
8 ! 3. Analyze Information
9 Router# show processes cpu sorted
10 Router# show memory
11 Router# show interfaces counters errors
12 Router# show buffers
13
14 ! 4. Eliminate Possibilities
15 Router# traceroute 8.8.8.8
16 Router# ping 8.8.8.8
17 Router# pathping 8.8.8.8
18
19 ! 5. Propose Hypothesis
20 ! 6. Test Hypothesis
21 Router# debug ip packet
22 Router# debug ip ospf events
23 Router# debug ip eigrp
24 Router# debug ip bgp updates
25
26 ! 7. Solve the Problem
```

```

27 Router# reload
28 Router# clear ip route *
29 Router# clear arp
30 Router# clear mac address-table
31
32 ! 8. Document Solution
33 Router# show running-config
34 Router# copy running-config startup-config

```

27.3 Common Troubleshooting Commands

```

1 ! Connectivity Testing
2 Router# ping 8.8.8.8
3 Router# ping 8.8.8.8 source Loopback0
4 Router# ping 8.8.8.8 size 1500 df-bit
5 Router# ping ipv6 2001:db8::1
6 Router# traceroute 8.8.8.8
7 Router# traceroute ipv6 2001:db8::1
8 Router# pathping 8.8.8.8
9
10 ! Packet Capture
11 Router# monitor capture CAP interface GigabitEthernet0/0 both
12 Router# monitor capture CAP match ipv4 any any
13 Router# monitor capture CAP start
14 ! ... wait for traffic ...
15 Router# monitor capture CAP stop
16 Router# monitor capture CAP export flash:capture.pcap
17
18 ! Performance Monitoring
19 Router# show processes cpu history
20 Router# show processes cpu sorted
21 Router# show processes memory
22 Router# show memory statistics
23 Router# show interfaces statistics
24 Router# show platform cpu packet statistics
25 Router# show platform hardware qfp active statistics drop

```

27.4 Debug Commands

```

1 ! Selective Debugging
2 Router# debug ip packet detail ! CAUTION: Very verbose!
3 Router# debug condition interface GigabitEthernet0/0
4 Router# debug condition ip 192.168.1.1
5 Router# debug ip ospf hello
6 Router# debug ip eigrp packets
7 Router# debug ip bgp updates
8 Router# debug ppp negotiation
9 Router# debug isdn q921
10 Router# debug dialer

```

```
11 Router# debug frame-relay lmi
12
13 ! Control Debug Output
14 Router# terminal monitor
15 Router# terminal no monitor
16 Router# undebug all
17 Router# no debug all
18 Router# debug sanity ! Check debug impact
19
20 ! Log Debug to Buffer
21 Router# logging buffered 100000 debugging
22 Router# show logging
```

28 Best Practices

28.1 Configuration Management

```
1 ! Configuration Archive Best Practices
2 Router(config)# archive
3 Router(config-archive)# path flash:archive-$h
4 Router(config-archive)# write-memory
5 Router(config-archive)# time-period 1440
6 Router(config-archive)# maximum 14
7 Router(config-archive)# exit
8
9 ! Automated Backups
10 Router(config)# kron policy-list BACKUP
11 Router(config-kron-policy)# cli write memory
12 Router(config-kron-policy)# cli copy running-config tftp
13 ://192.168.100.10/backups/$h-config
14 Router(config-kron-policy)# exit
15
16 Router(config)# kron occurrence DAILY-BACKUP at 2:00 recurring
17 Router(config-kron-occurrence)# policy-list BACKUP
18 Router(config-kron-occurrence)# exit
```

28.2 Security Hardening

```
1 ! Security Checklist
2 ! 1. Disable Unused Services
3 no service pad
4 no ip source-route
5 no ip bootp server
6 no service tcp-small-servers
7 no service udp-small-servers
8 no ip http server
9 no ip http secure-server
10 no cdp run
```

```

11 no lldp run
12
13 ! 2. Enable Password Encryption
14 service password-encryption
15 enable secret [strong-password]
16 security passwords min-length 12
17
18 ! 3. Configure AAA
19 aaa new-model
20 aaa authentication login default local
21 aaa authorization exec default local
22 username admin privilege 15 secret [hash]
23
24 ! 4. Enable SSH
25 ip domain-name company.com
26 crypto key generate rsa modulus 2048
27 ip ssh version 2
28 ip ssh time-out 60
29 ip ssh authentication-retries 2
30
31 ! 5. Configure Logging
32 logging host 192.168.100.10
33 logging trap debugging
34 logging source-interface Loopback0
35
36 ! 6. Implement ACLs
37 ! 7. Enable NTP
38 ! 8. Regular Updates

```

28.3 Performance Optimization

```

1 ! CPU Optimization
2 Router(config)# process cpu threshold type total rising 80
   interval 5 falling 70 interval 5
3 Router(config)# ip cef
4 Router(config)# mls cef
5
6 ! Memory Optimization
7 Router(config)# memory reserve critical 4096
8 Router(config)# buffers small permanent 100
9 Router(config)# buffers middle permanent 100
10 Router(config)# buffers big permanent 100
11
12 ! Interface Optimization
13 Router(config-if)# no ip route-cache cef
14 Router(config-if)# ip tcp adjust-mss 1360
15 Router(config-if)# ip mtu 1500
16 Router(config-if)# keepalive 10
17
18 ! Routing Optimization

```

```
19 Router(config-router)# timers basic 5 15 5 40
20 Router(config-router)# maximum-paths 4
21 Router(config-router)# traffic-share balanced
```

28.4 Documentation Standards

```
1  ! Configuration Header
2  !
3  ! Device: R1-CORE
4  ! Location: Data Center Rack A1
5  ! Purpose: Core Router for Company Network
6  ! Contact: Network Operations (noc@company.com)
7  ! Last Modified: 2024-01-01 by John Doe
8  ! Change Ticket: NET-2024-001
9  !
10 ! Interface Descriptions
11 interface GigabitEthernet0/0
12     description Uplink to ISP1 - Circuit ID: ISP-001
13 interface GigabitEthernet0/1
14     description Uplink to ISP2 - Circuit ID: ISP-002
15 interface Loopback0
16     description Router ID and Management
17 !
18 ! VLAN Documentation
19 vlan 10
20     name SALES
21     description Sales Department VLAN
22 !
23 ! ACL Documentation
24 ip access-list extended INTERNET-ACCESS
25     remark Allow HTTP/HTTPS from Internal Network
26     remark Allow DNS queries
27     remark Deny and log all other traffic
28 !
29 ! Routing Documentation
30 router ospf 100
31     ! Area 0 - Backbone Area
32     ! Area 1 - Branch Offices (Stub)
33     ! Area 2 - Data Center (NSSA)
34 !
35 ! BGP Documentation
36 router bgp 65001
37     ! eBGP with ISP1 (AS 65002)
38     ! eBGP with ISP2 (AS 65003)
39     ! iBGP with Internal Routers
40 !
```

29 Common Issues & Fixes

29.1 Interface Issues

```
1 ! Interface Won't Come Up
2 ! 1. Check Physical Connection
3 Router# show interfaces gigabitEthernet 0/0
4 ! Look for: administratively down, line protocol down
5
6 ! 2. Enable Interface
7 Router(config)# interface gigabitEthernet 0/0
8 Router(config-if)# no shutdown
9
10 ! 3. Check Speed/Duplex
11 Router# show interfaces gigabitEthernet 0/0 status
12 Router(config-if)# speed 1000
13 Router(config-if)# duplex full
14 Router(config-if)# negotiation auto
15
16 ! 4. Check VLAN Configuration
17 Router# show interfaces gigabitEthernet 0/0 switchport
18 Router(config-if)# switchport mode access
19 Router(config-if)# switchport access vlan 10
20
21 ! 5. Check Errors
22 Router# show interfaces gigabitEthernet 0/0 counters errors
23 Router# clear counters gigabitEthernet 0/0
```

29.2 Routing Issues

```
1 ! OSPF Neighbor Not Forming
2 ! 1. Check Connectivity
3 Router# ping <neighbor-ip>
4
5 ! 2. Check OSPF Parameters
6 Router# show ip ospf interface gigabitEthernet 0/0
7 ! Verify: Area, Hello/Dead Timers, Authentication
8
9 ! 3. Check Network Type
10 Router# show ip ospf interface gigabitEthernet 0/0
11 Router(config-if)# ip ospf network point-to-point
12
13 ! 4. Check ACLs/Firewall
14 Router# show ip access-lists
15 Router# show zone-pair security
16
17 ! 5. Debug OSPF
18 Router# debug ip ospf adj
19 Router# debug ip ospf hello
```

```
20
21 ! BGP Peer Not Coming Up
22 ! 1. Check TCP Connectivity
23 Router# telnet <peer-ip> 179
24
25 ! 2. Check BGP Parameters
26 Router# show ip bgp neighbors
27 ! Verify: AS Number, Authentication, Timers
28
29 ! 3. Check Route to Peer
30 Router# show ip route <peer-ip>
31 Router(config)# ip route <peer-ip> 255.255.255.255 <next-hop>
32
33 ! 4. Debug BGP
34 Router# debug ip bgp
35 Router# debug ip tcp transactions
```

29.3 VPN Issues

```
1 ! IPsec VPN Not Establishing
2 ! 1. Check Connectivity
3 Router# ping <peer-ip>
4
5 ! 2. Check IKE Policy
6 Router# show crypto isakmp policy
7 ! Verify: Encryption, Hash, DH Group, Lifetime
8
9 ! 3. Check Transform Sets
10 Router# show crypto ipsec transform-set
11 ! Verify: Encryption, Hash, Mode
12
13 ! 4. Check Interesting Traffic
14 Router# show crypto map
15 Router# show access-lists
16
17 ! 5. Check NAT/Traversal
18 Router(config-if)# ip nat inside
19 Router(config-if)# ip nat outside
20 Router# debug crypto isakmp
21 Router# debug crypto ipsec
```

29.4 Switching Issues

```
1 ! VLAN Issues
2 ! 1. Check VLAN Configuration
3 Switch# show vlan brief
4 Switch# show vlan id 10
5
6 ! 2. Check Trunk Configuration
```

```

7 Switch# show interfaces trunk
8 Switch# show interfaces gigabitEthernet 1/0/24 switchport
9
10 ! 3. Check STP
11 Switch# show spanning-tree vlan 10
12 Switch# show spanning-tree inconsistentports
13
14 ! 4. Check VTP
15 Switch# show vtp status
16 Switch# show vtp password
17
18 ! EtherChannel Issues
19 ! 1. Check Channel Group
20 Switch# show etherchannel summary
21 Switch# show etherchannel port-channel
22
23 ! 2. Check Protocol
24 Switch# show lacp neighbor
25 Switch# show pagp neighbor
26
27 ! 3. Check Configuration Consistency
28 Switch# show running-config interface port-channel 1
29 Switch# show running-config interface gigabitEthernet 1/0/1
30 Switch# show running-config interface gigabitEthernet 1/0/2

```

29.5 Performance Issues

```

1 ! High CPU Utilization
2 ! 1. Identify Process
3 Router# show processes cpu sorted
4 Router# show processes cpu history
5
6 ! 2. Check for Broadcast Storms
7 Router# show interfaces counters broadcast
8 Switch# show storm-control
9
10 ! 3. Check for Loops
11 Router# show spanning-tree
12 Switch# show spanning-tree inconsistentports
13
14 ! 4. Check Routing Protocols
15 Router# show ip ospf neighbor
16 Router# show ip bgp summary
17 Router# debug ip routing
18
19 ! High Memory Usage
20 Router# show memory
21 Router# show memory statistics
22 Router# show buffers
23 Router# show processes memory

```